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Lugg

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(54) **LOCKING MECHANISM USED WITH A FLAT WEBBING**

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This patent is subject to a terminal disclaimer.

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(60) Provisional application No. 63/155,739, filed on Mar. 3, 2021.

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A01K 27/00 (2006.01)

(52) **U.S. Cl.**
CPC **A01K 27/005** (2013.01); **A01K 27/003** (2013.01)

(58) **Field of Classification Search**
CPC A01K 27/001; A01K 27/003; A01K 27/27002; A01K 27/005; A44B 11/2515; A44B 11/2519; A44B 11/2592
See application file for complete search history.

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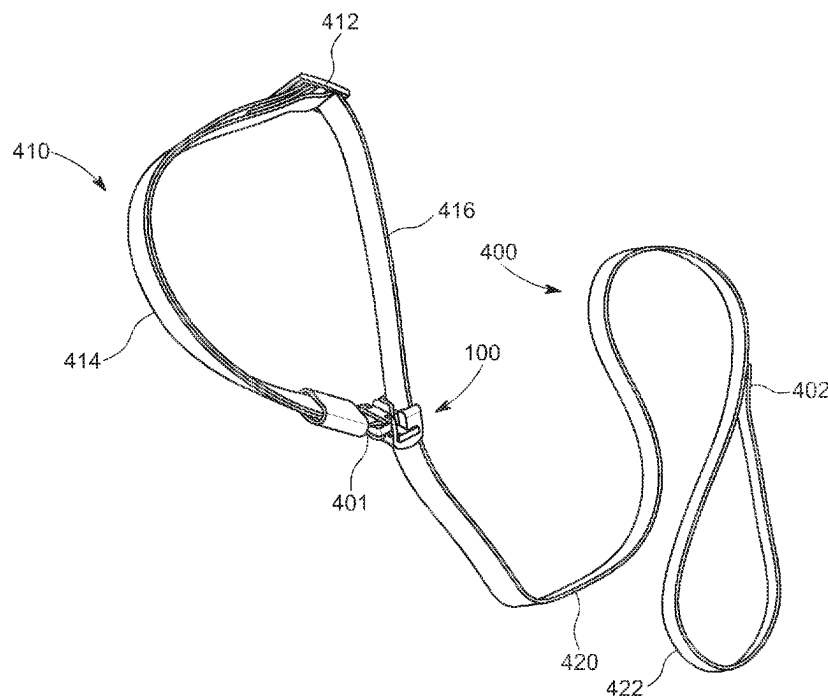
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(57) **ABSTRACT**

The present invention is a locking device that can be integrated onto a leash, and which provides a solution to the problem of slip-on collars over tightening or being inconvenient to use. The locking device comprises of a main body, a lock piece, a hook, and one or more arms. The one or more arms are connected within the main body to engage with the hook and the lock piece to lock a width of a collar portion which is formed from the pieces integrated onto the leash. The main body is configured to slide on the leash, and the lock piece is fixed onto the leash and when the lock piece slides into the main body, the main body is prevented from sliding. The hook is connected to a first end of the leash and when inserted into the main body forms the collar portion.

20 Claims, 23 Drawing Sheets



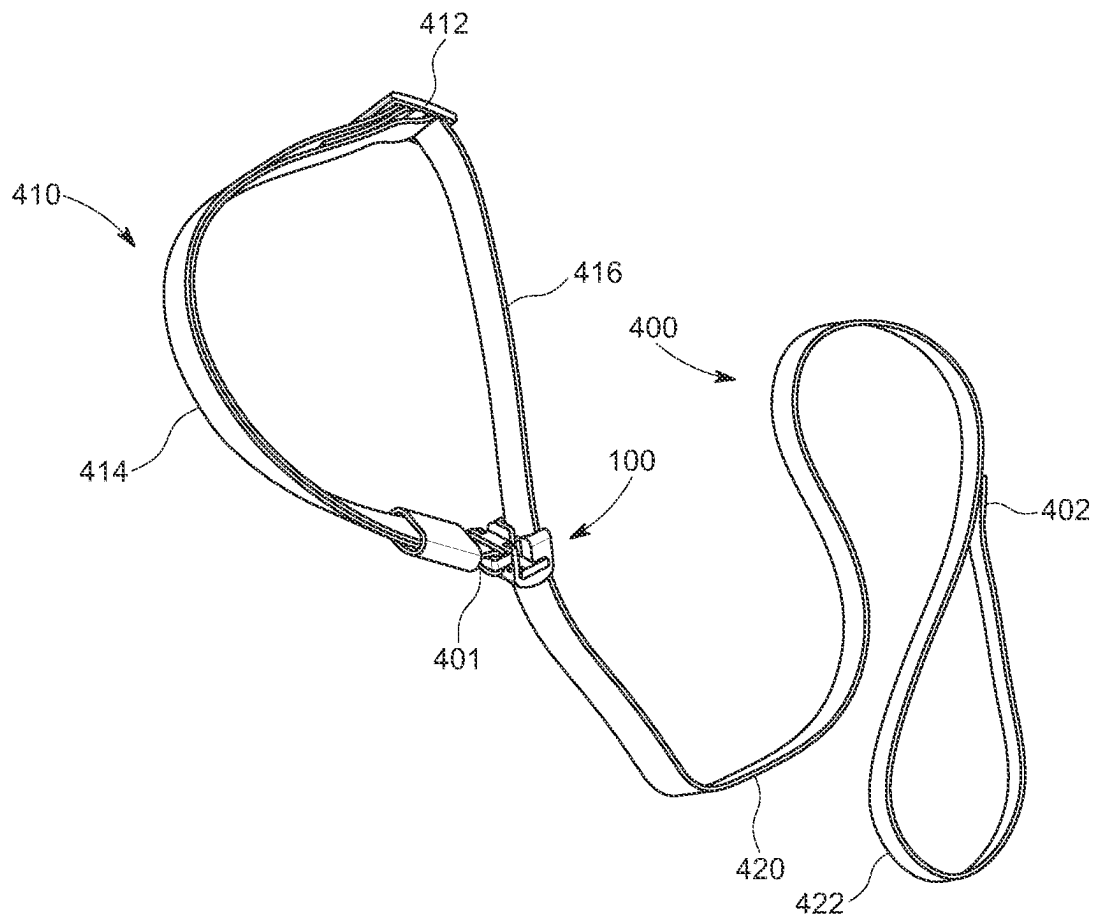


FIG. 1

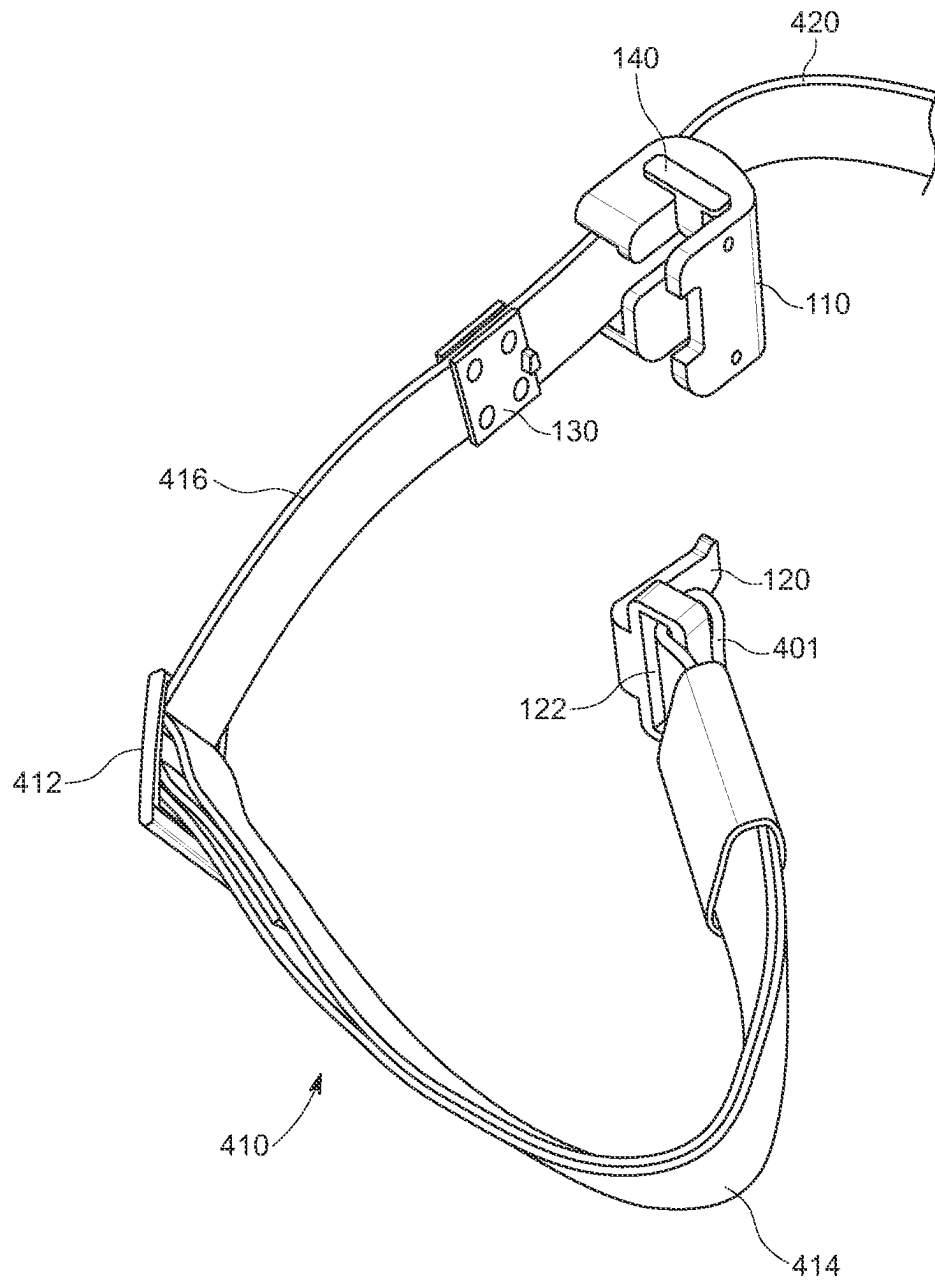


FIG. 2

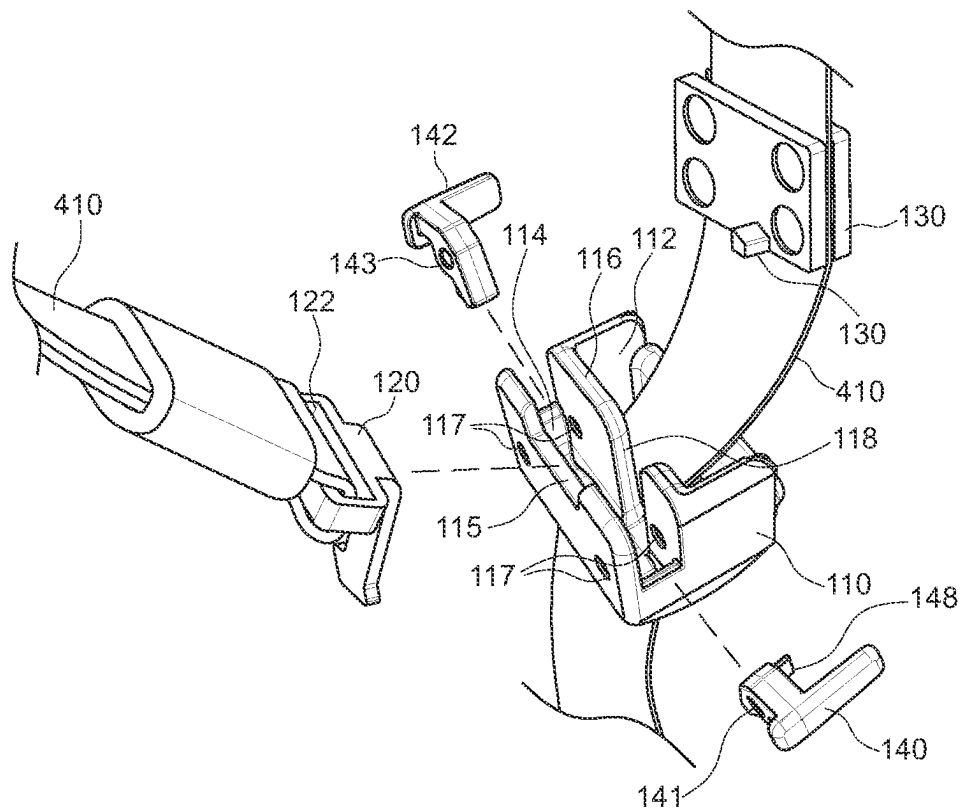


FIG. 3A

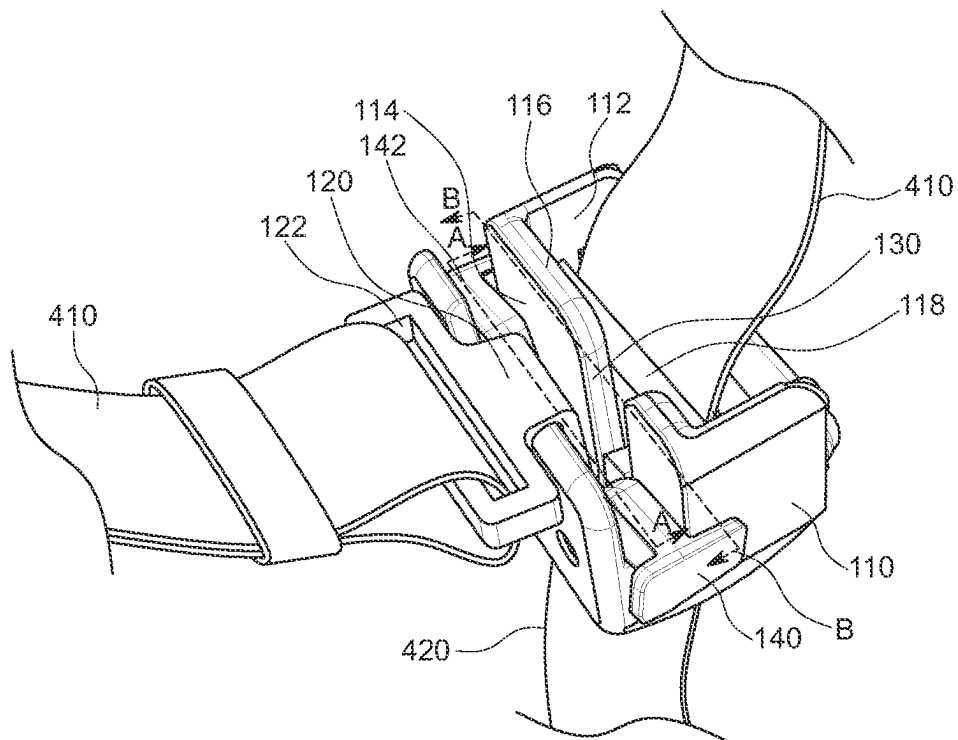


FIG. 3B

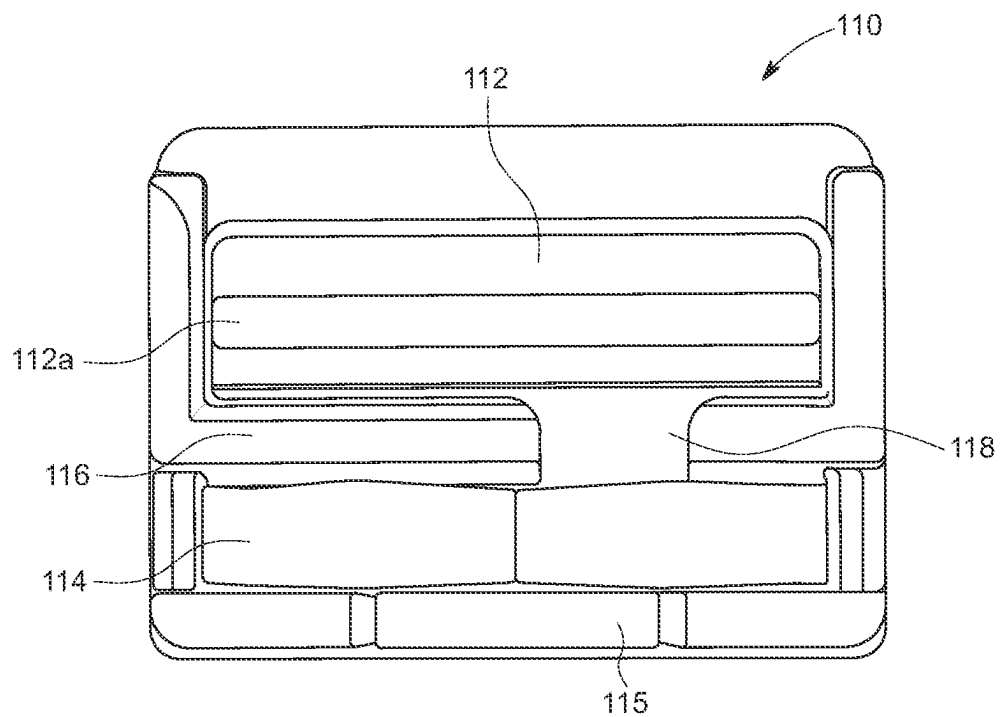


FIG. 4

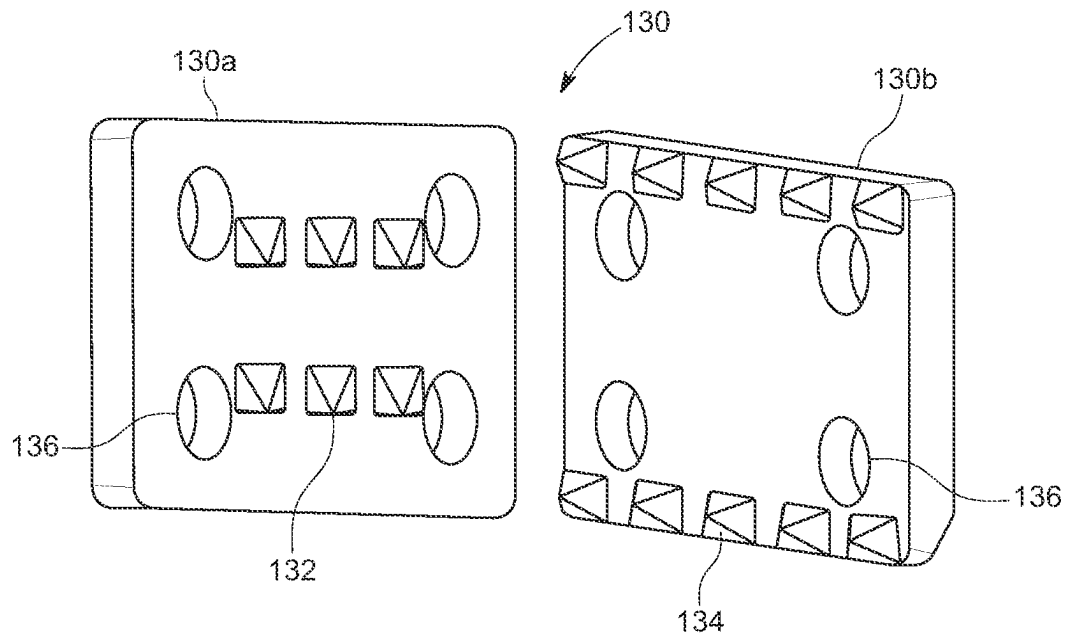


FIG. 5A

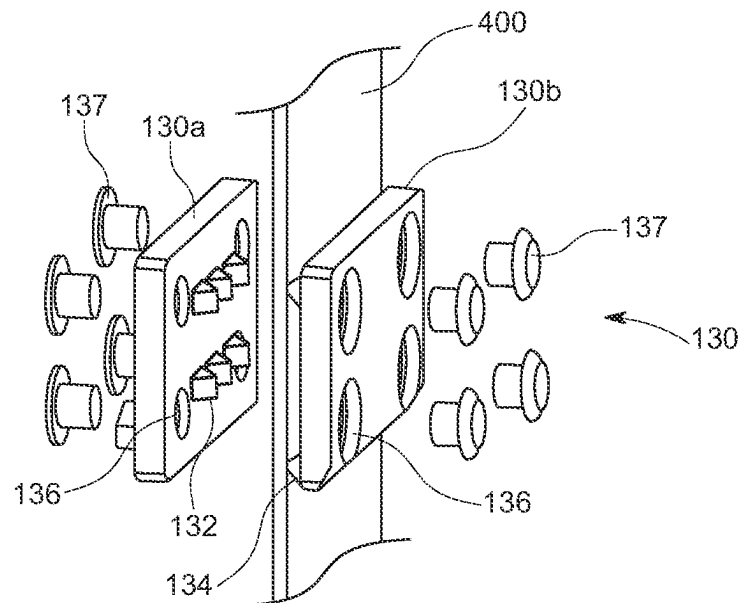


FIG. 5B

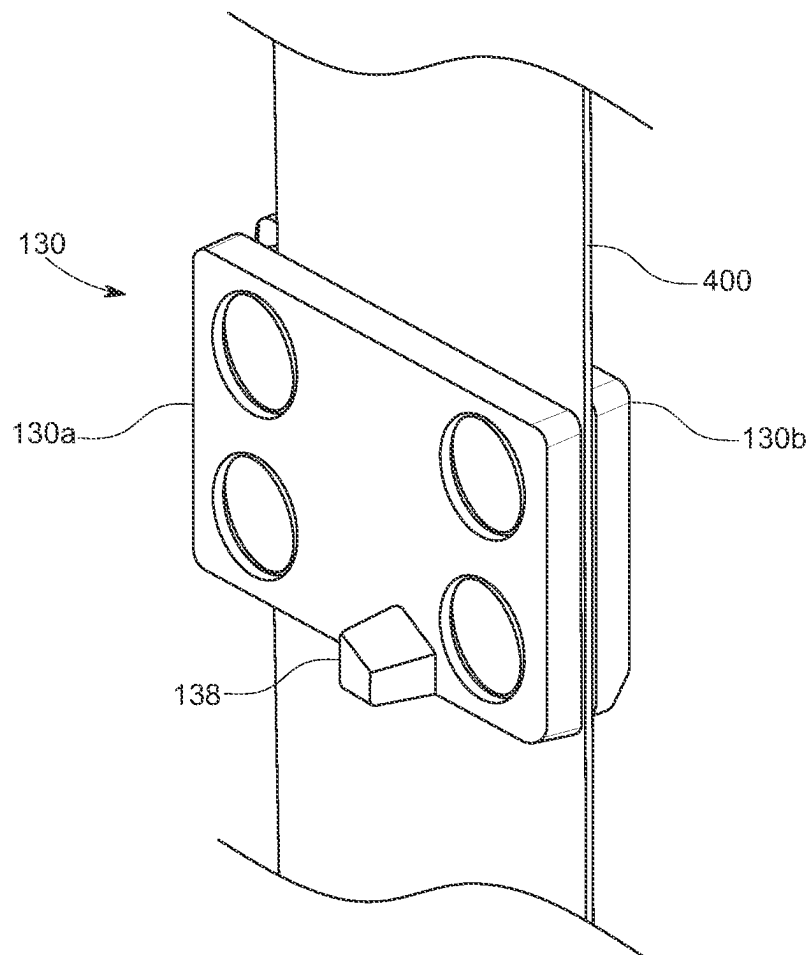


FIG. 5C

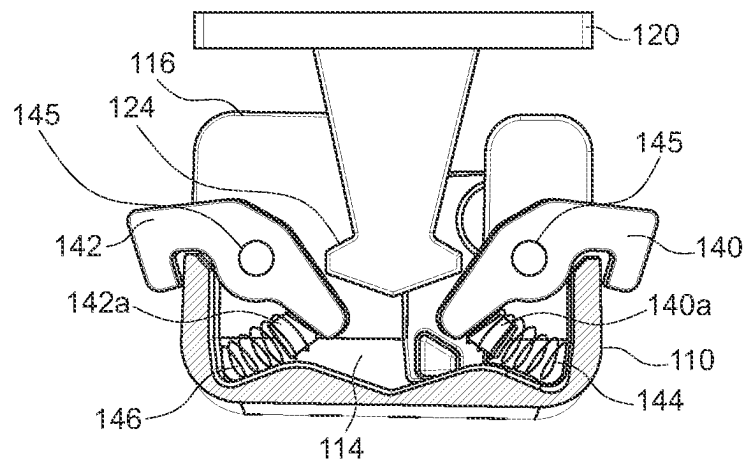


FIG. 6A

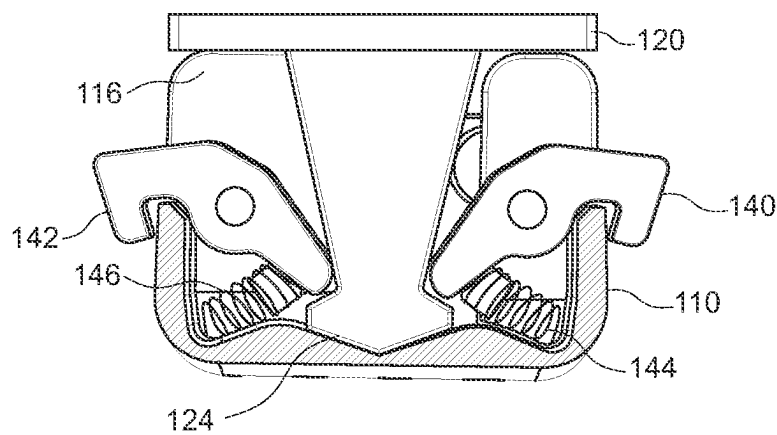


FIG. 6B

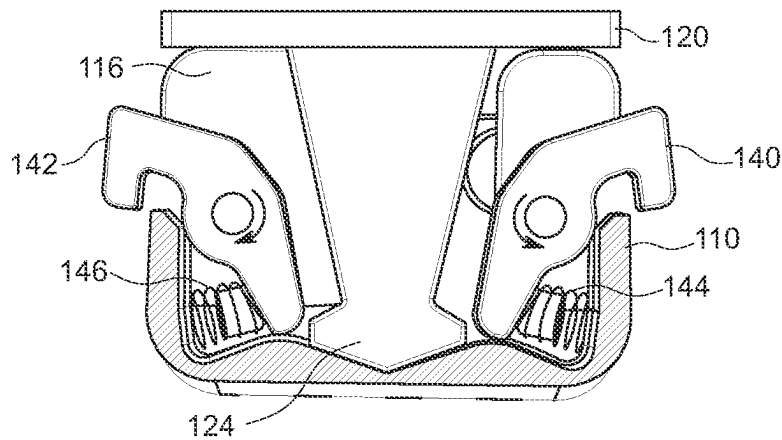


FIG. 6C

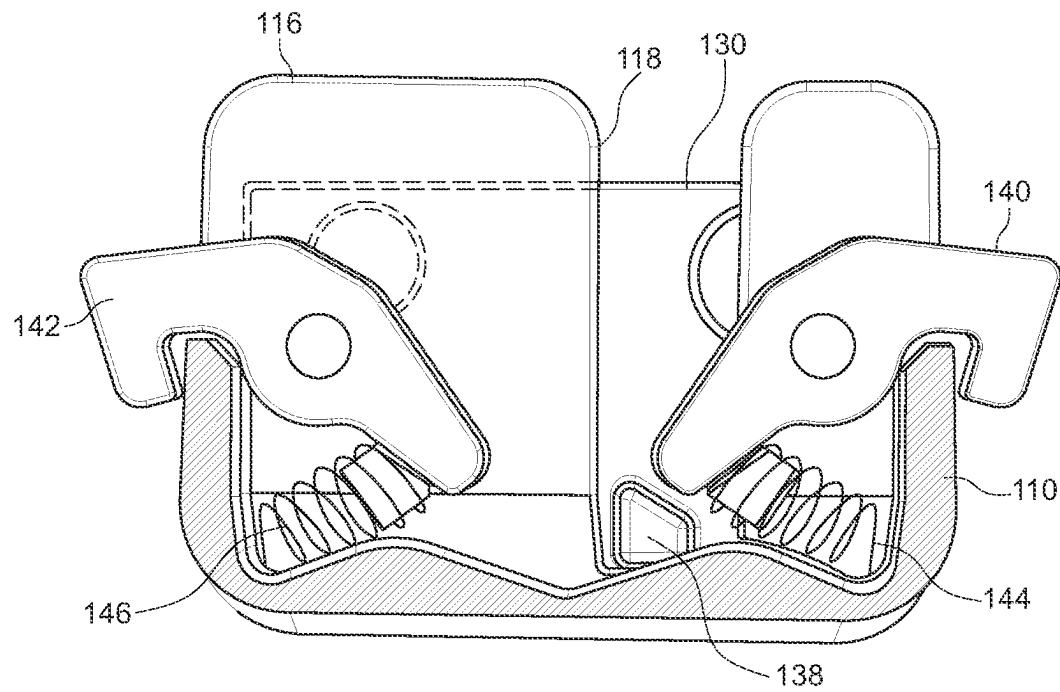


FIG. 7A

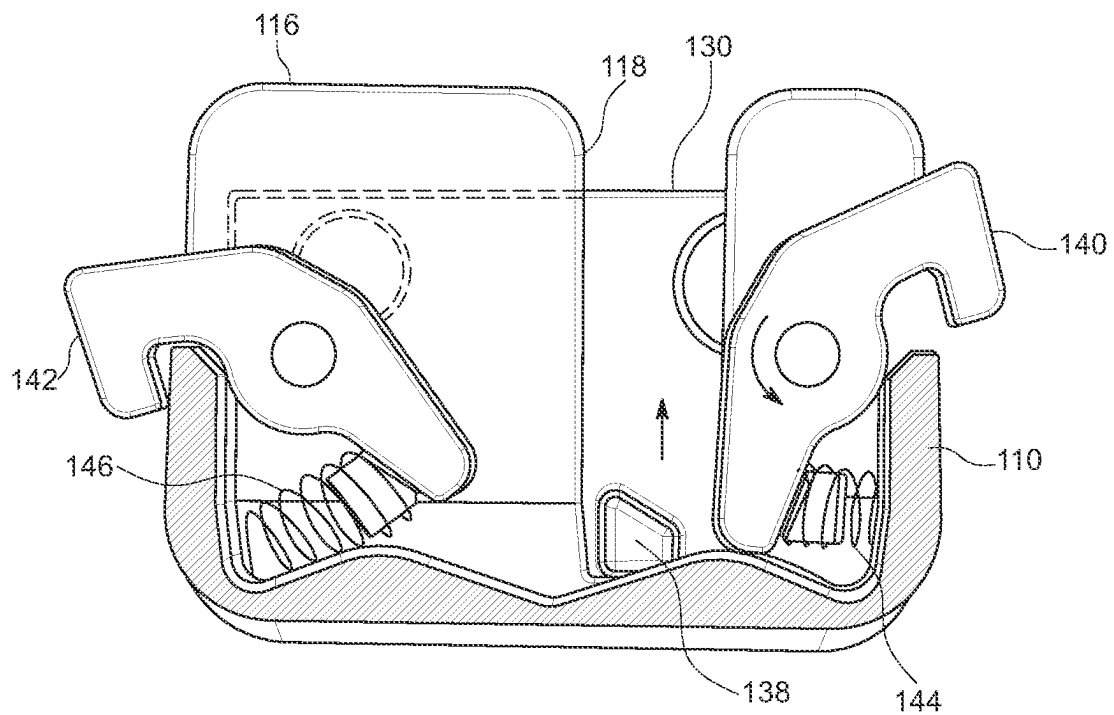


FIG. 7B

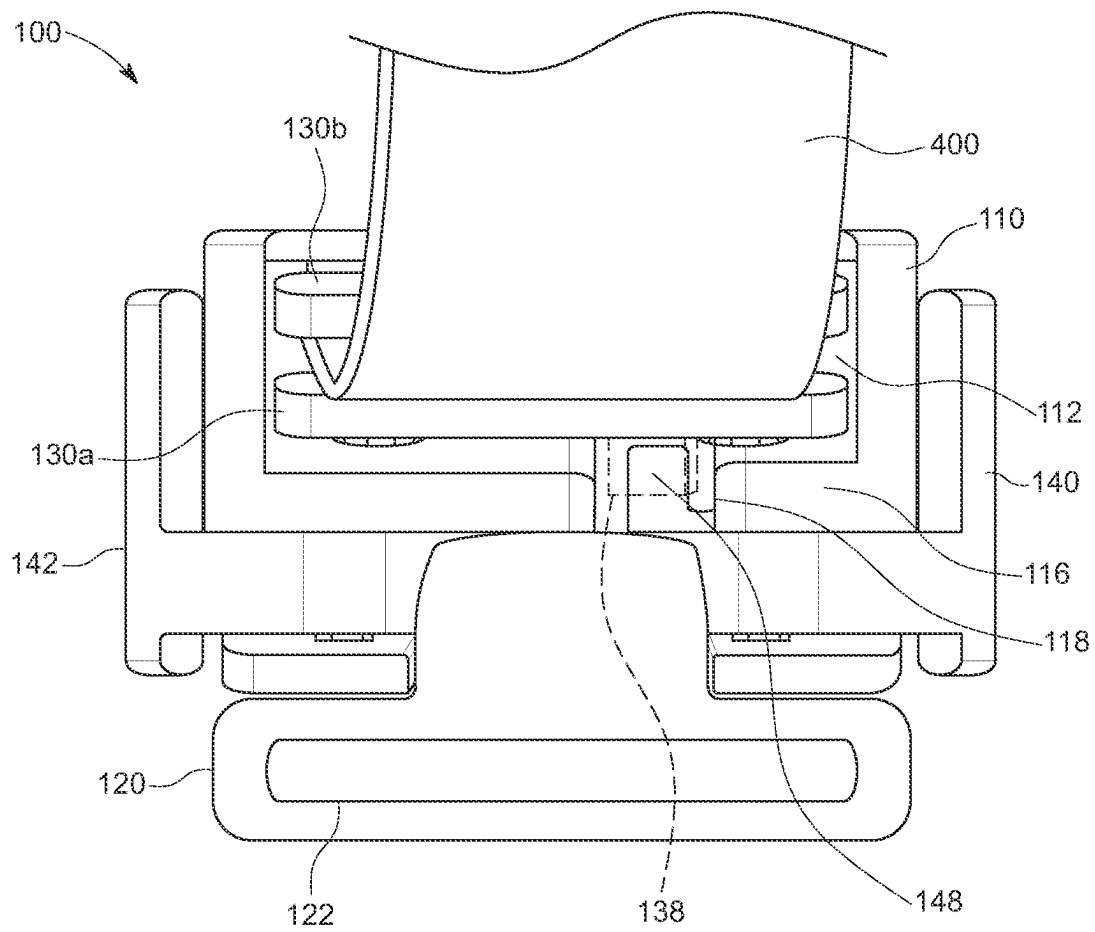


FIG. 8

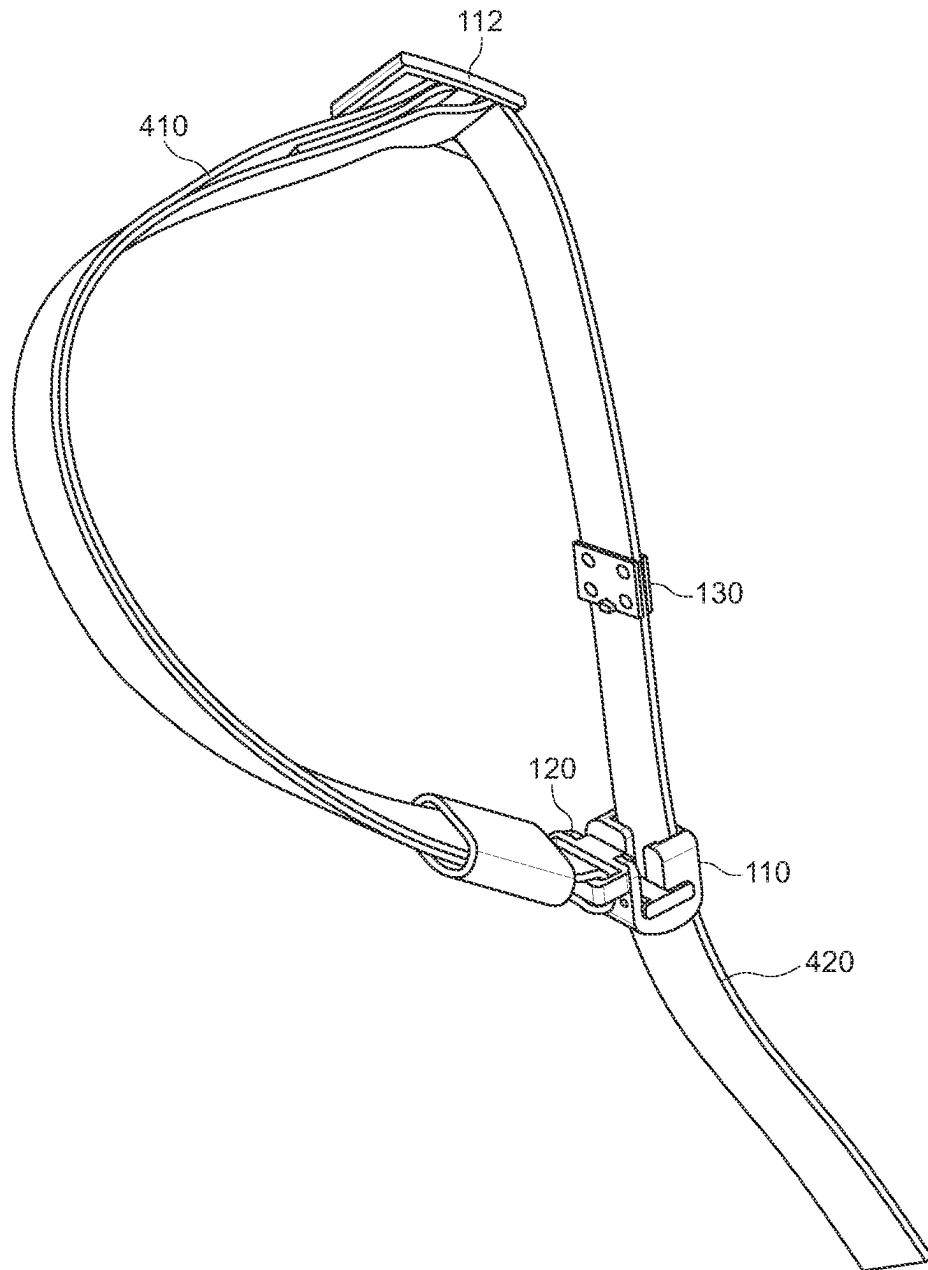


FIG. 9A

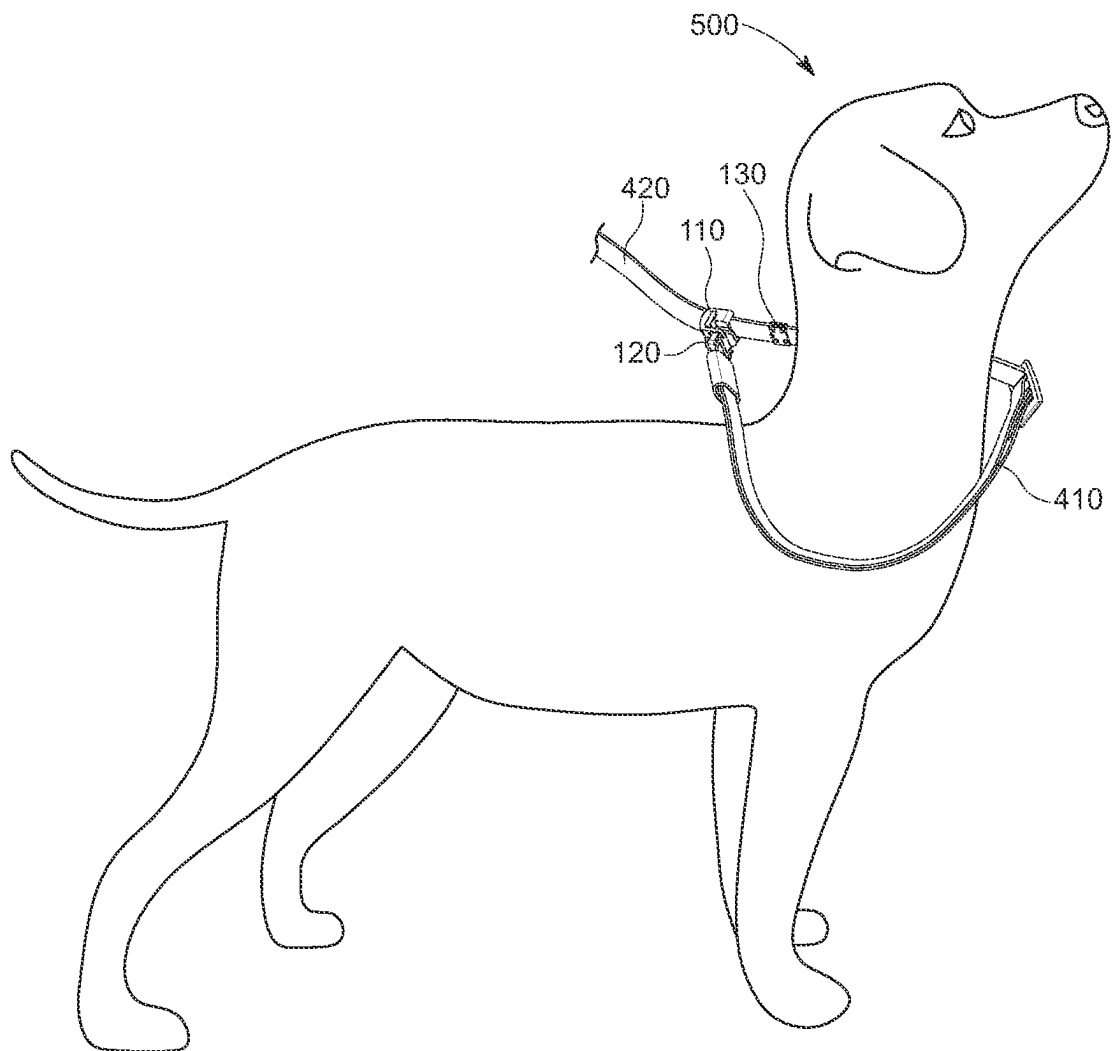


FIG. 9B

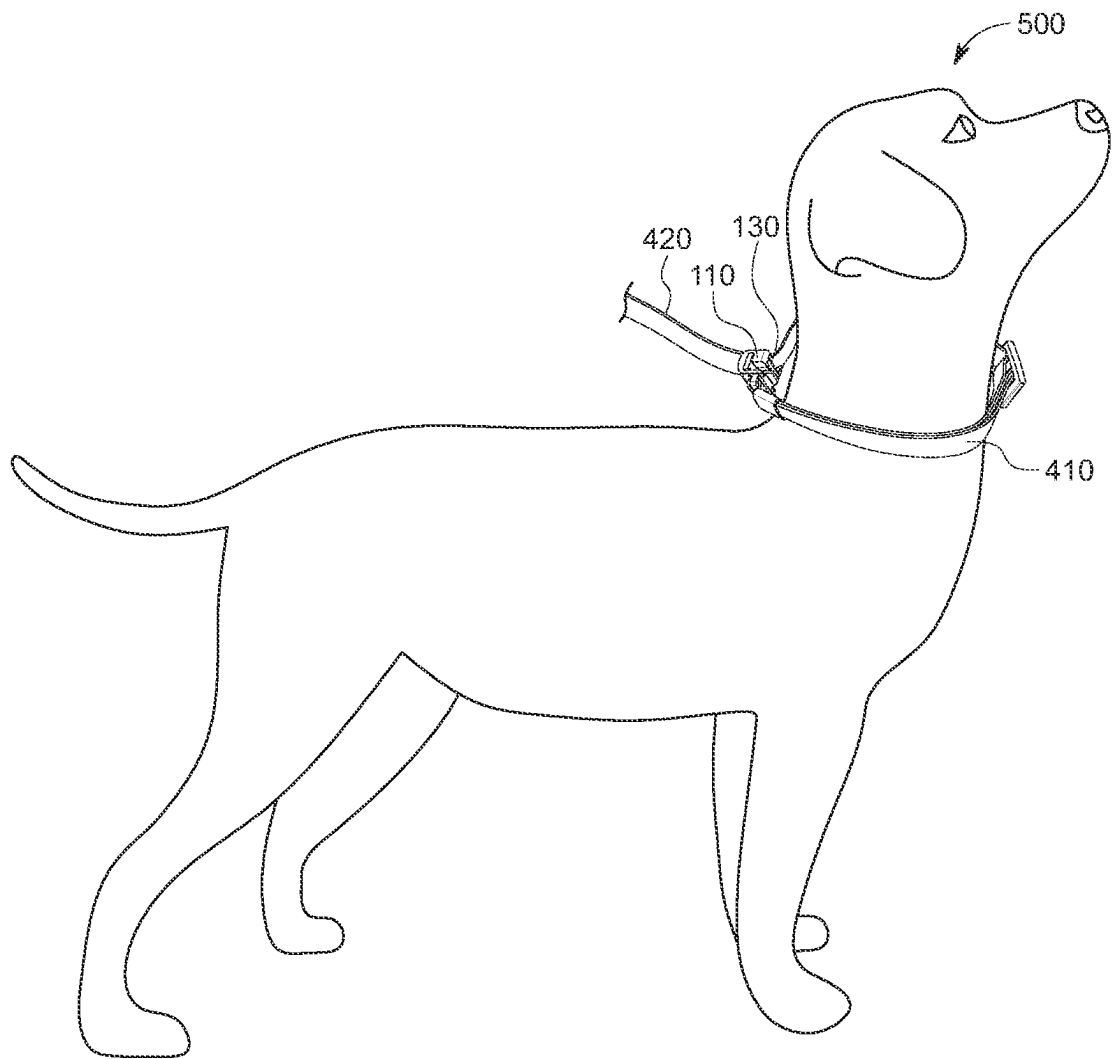


FIG. 9C

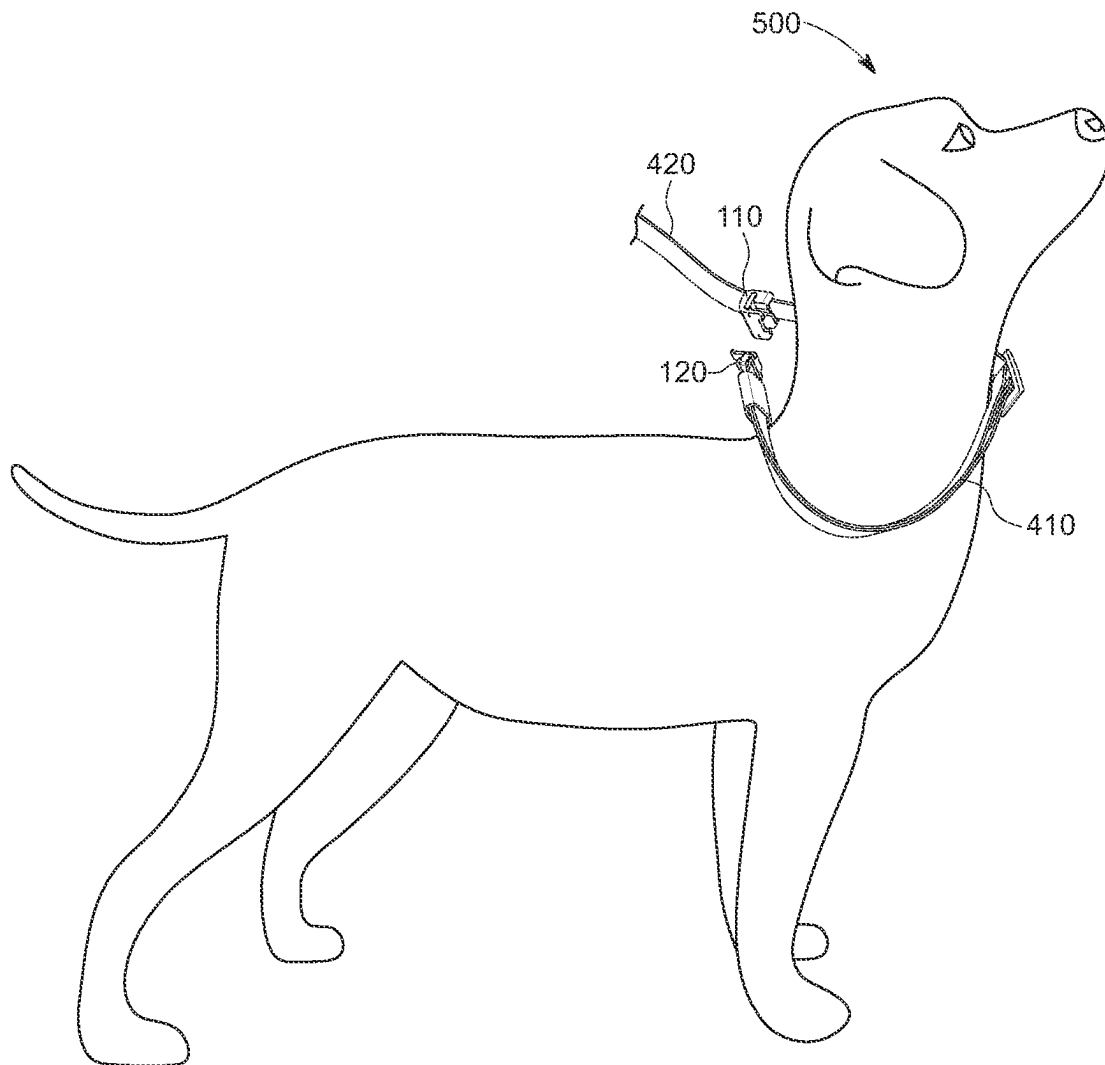


FIG. 9D

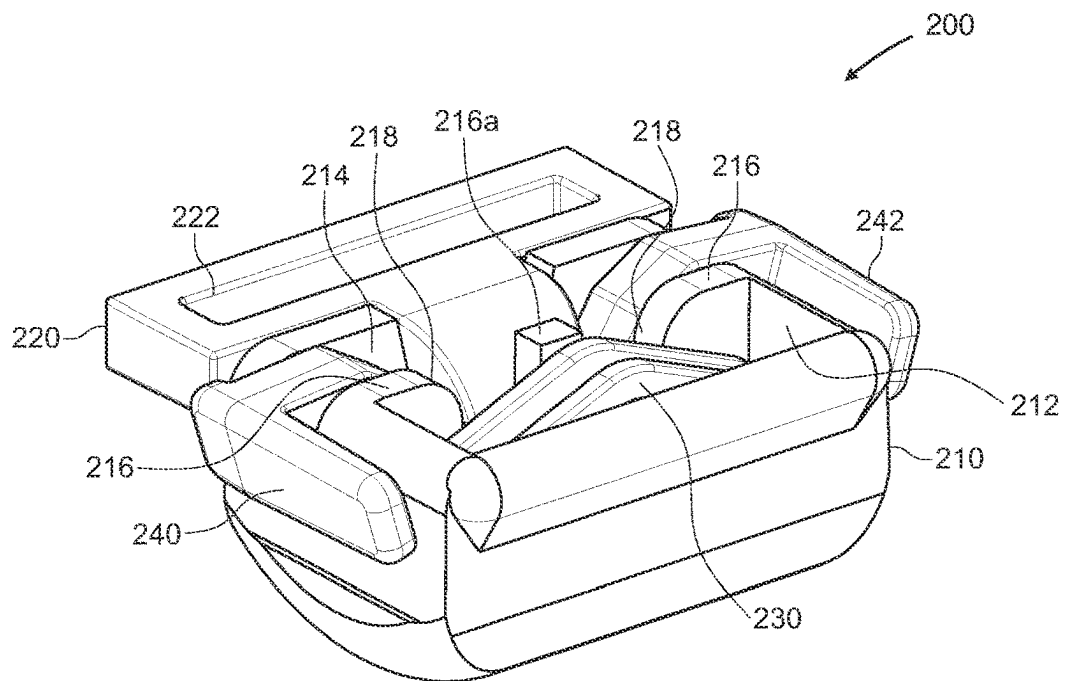


FIG. 10A

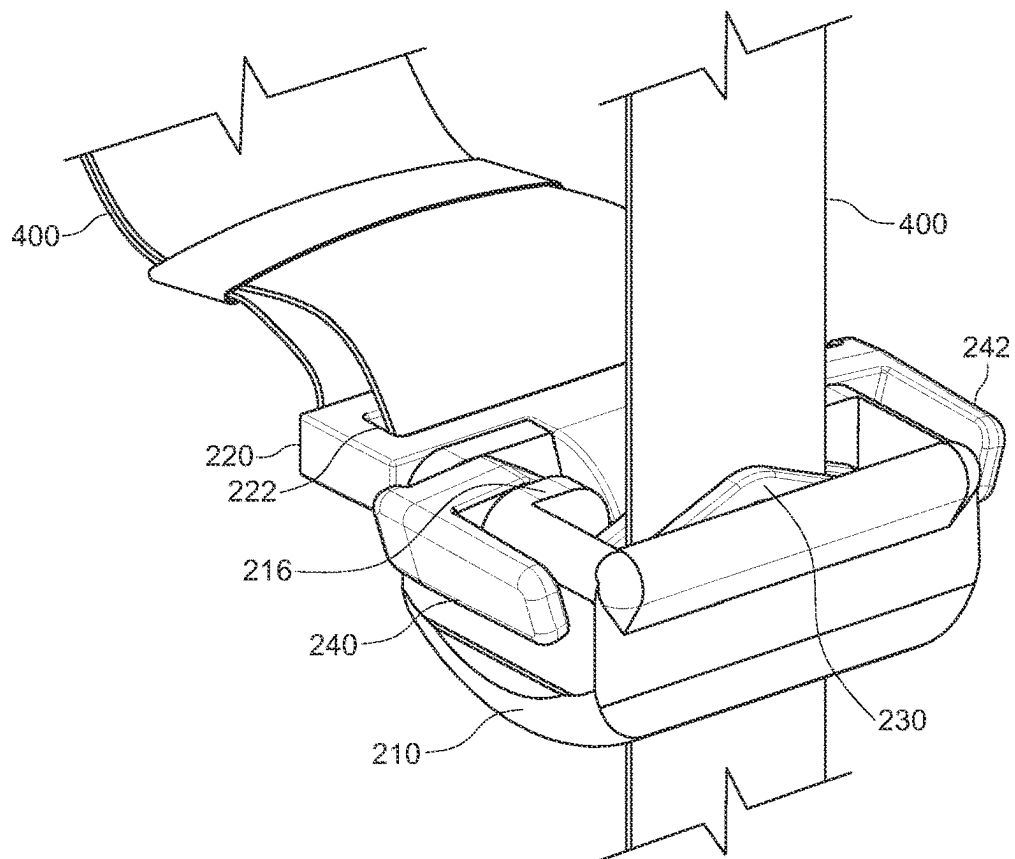


FIG. 10B

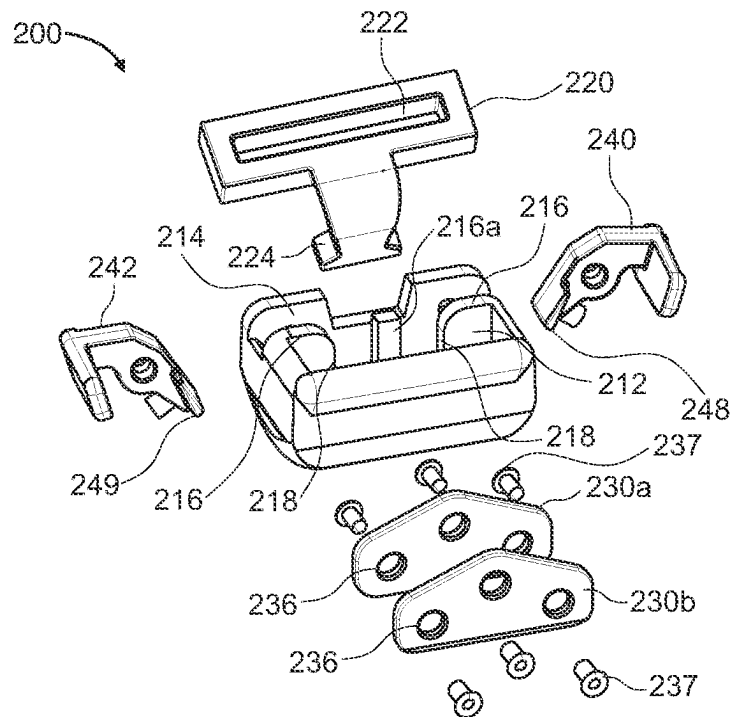


FIG. 11

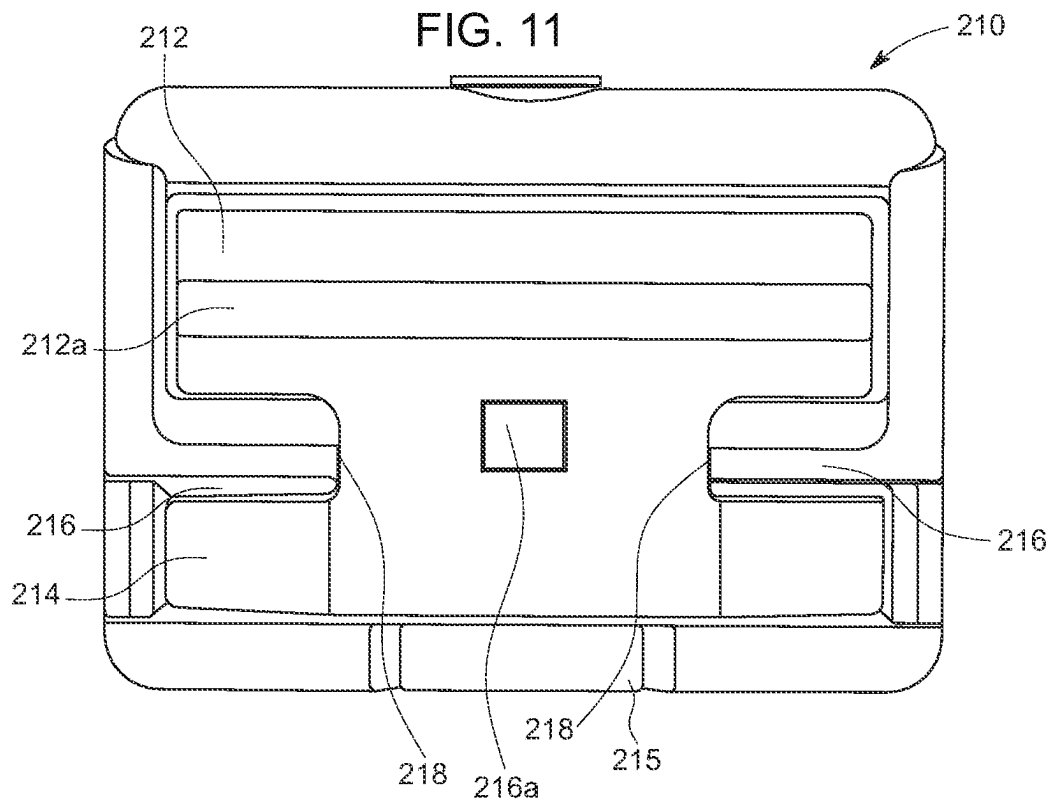


FIG. 12

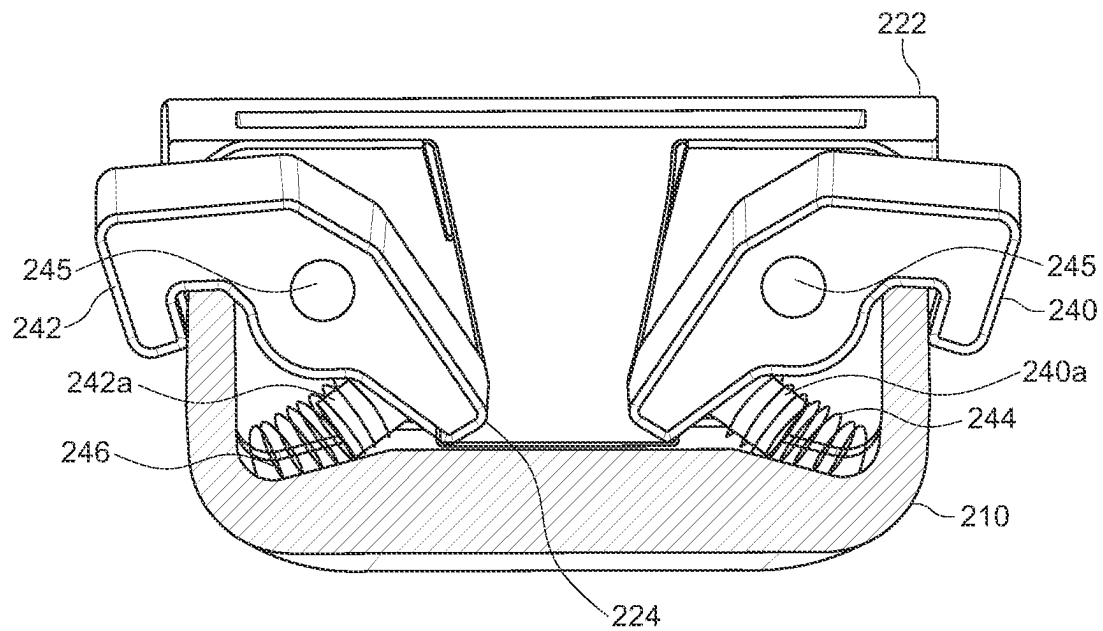


FIG. 13

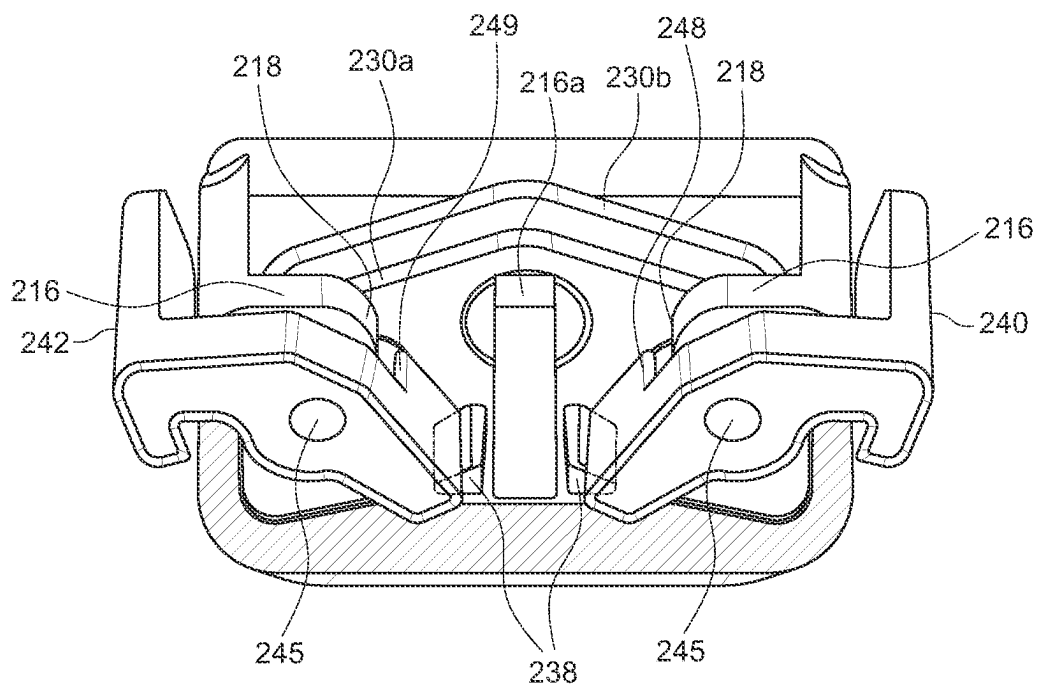


FIG. 14

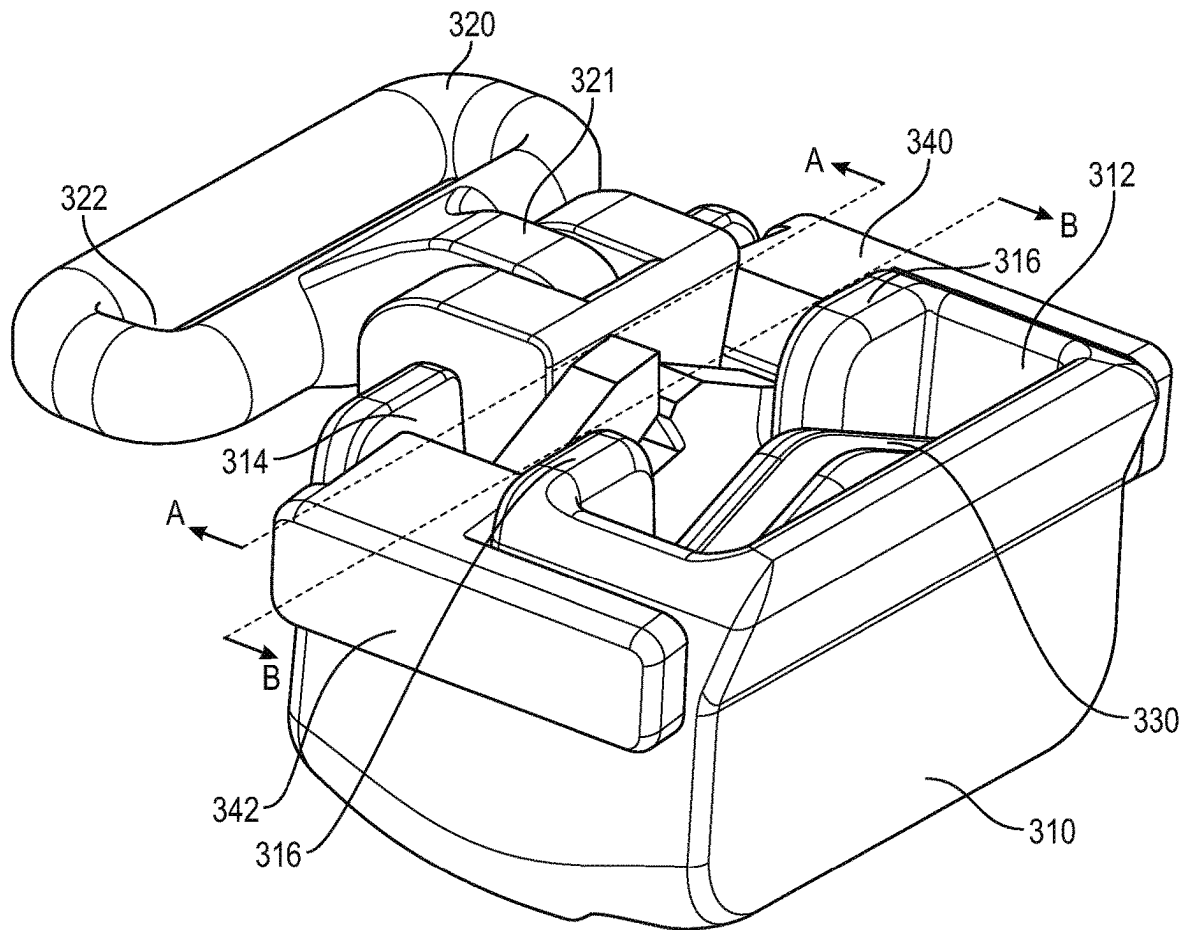


FIG. 15

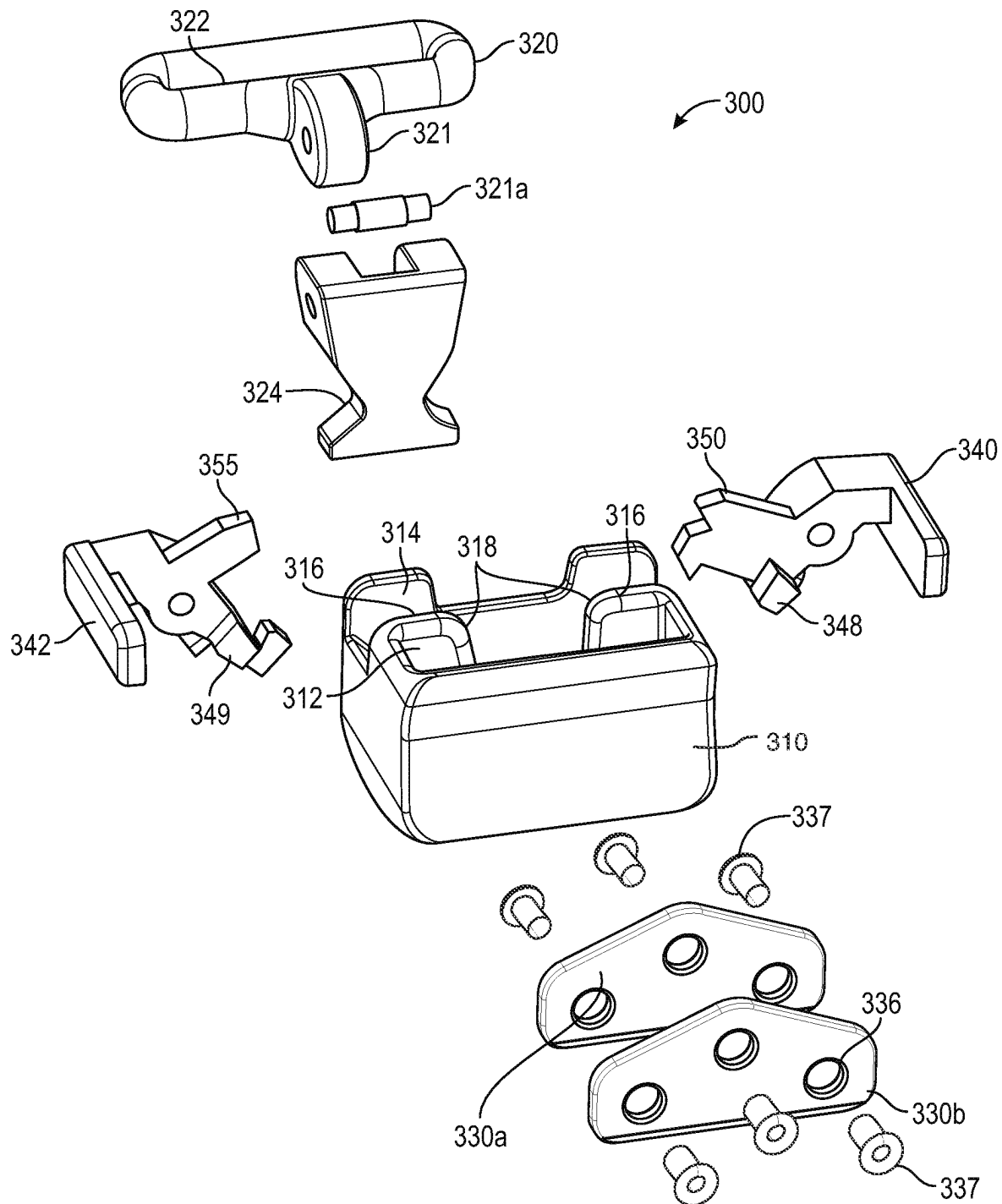


FIG. 16

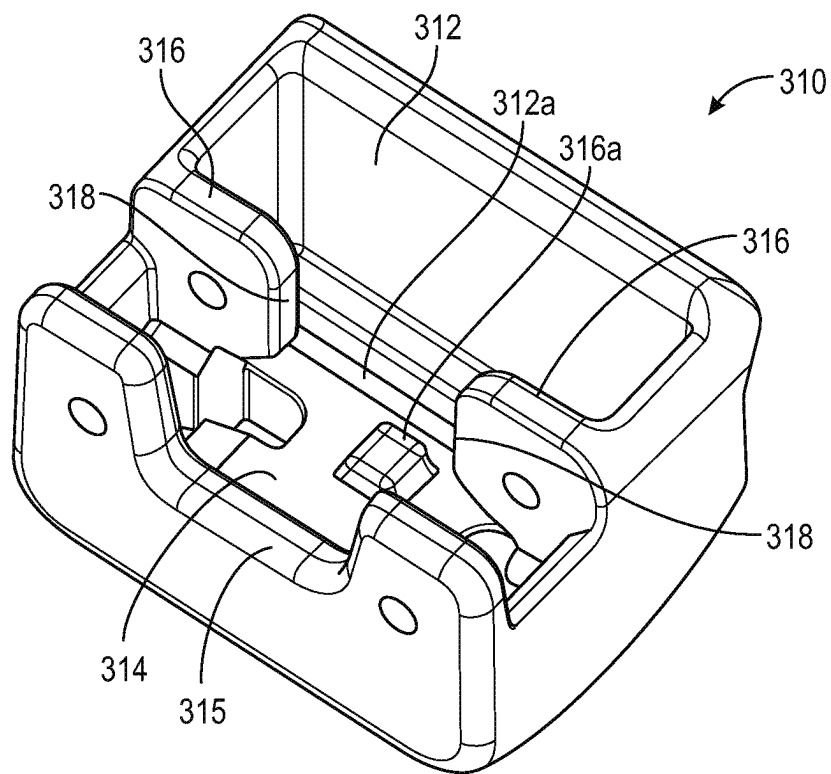


FIG. 17

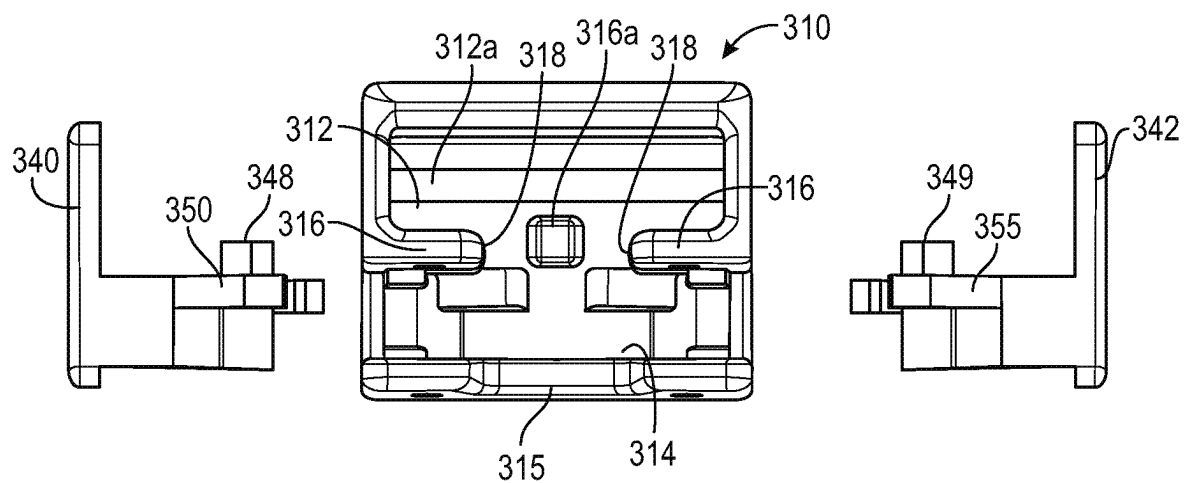


FIG. 18

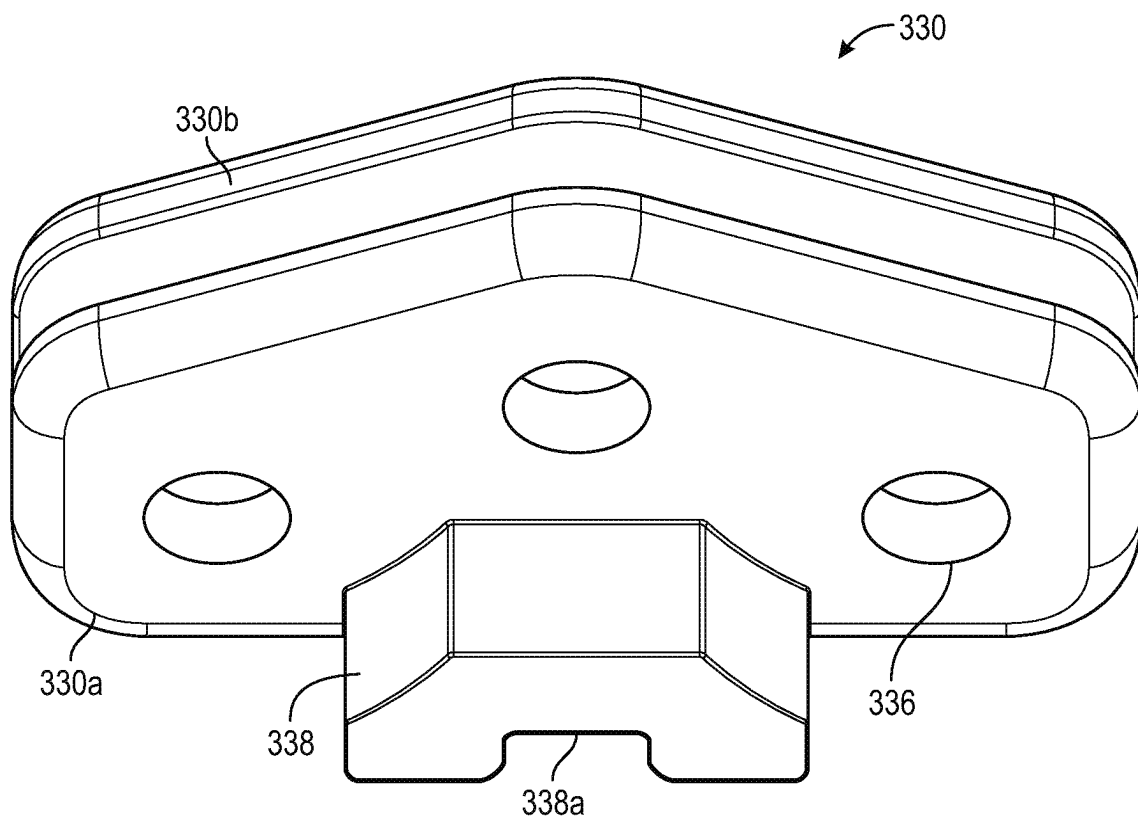


FIG. 19

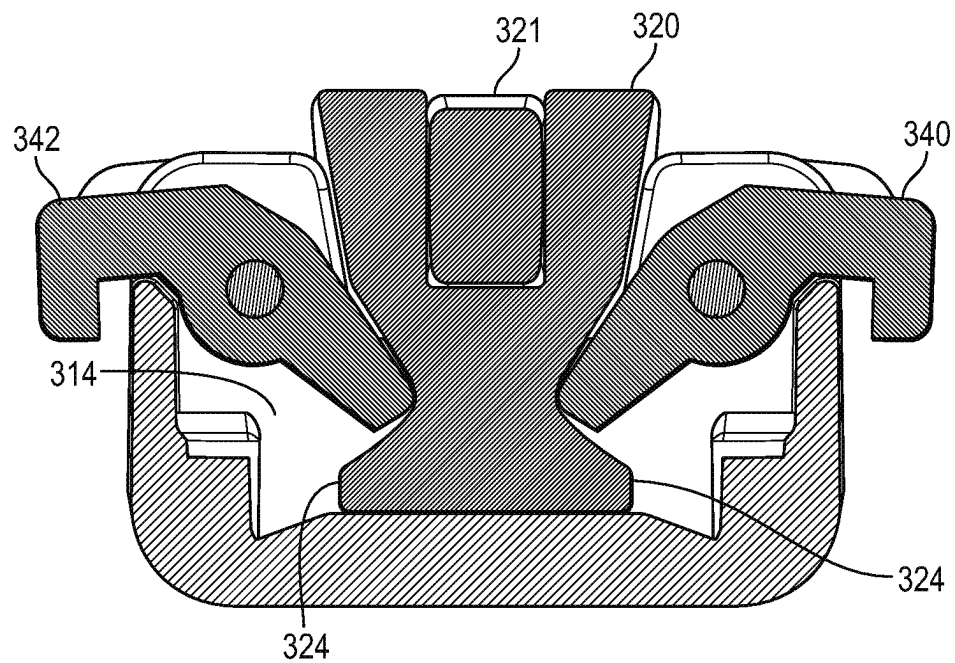


FIG. 20

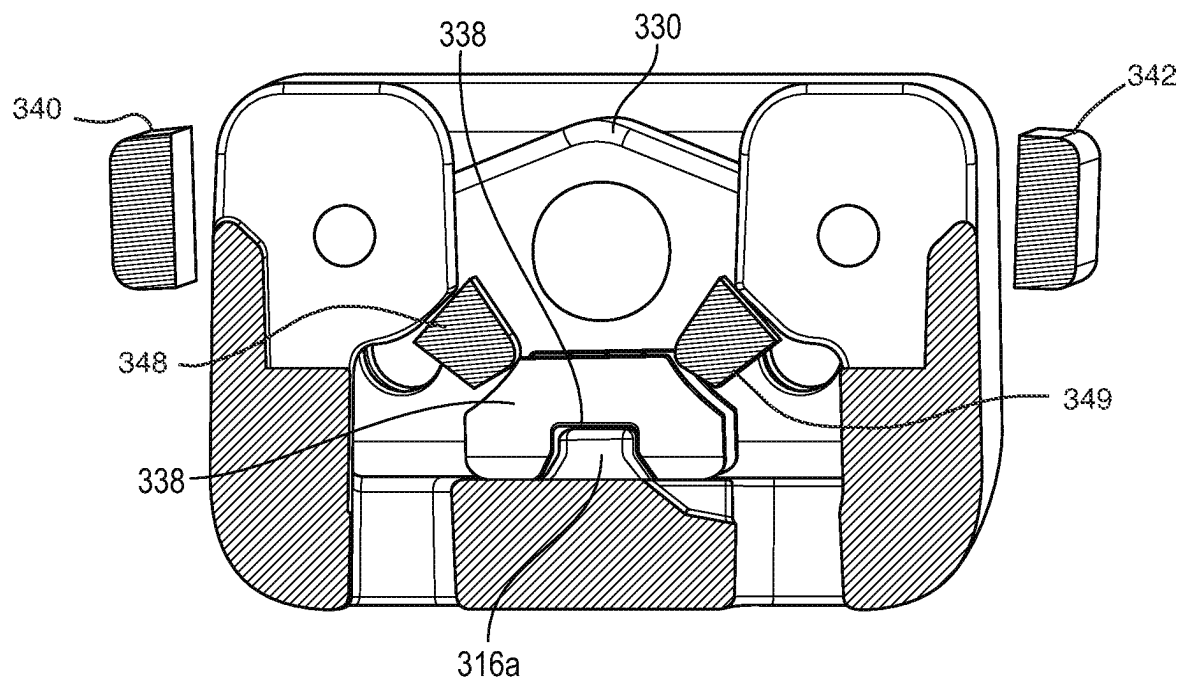


FIG. 21

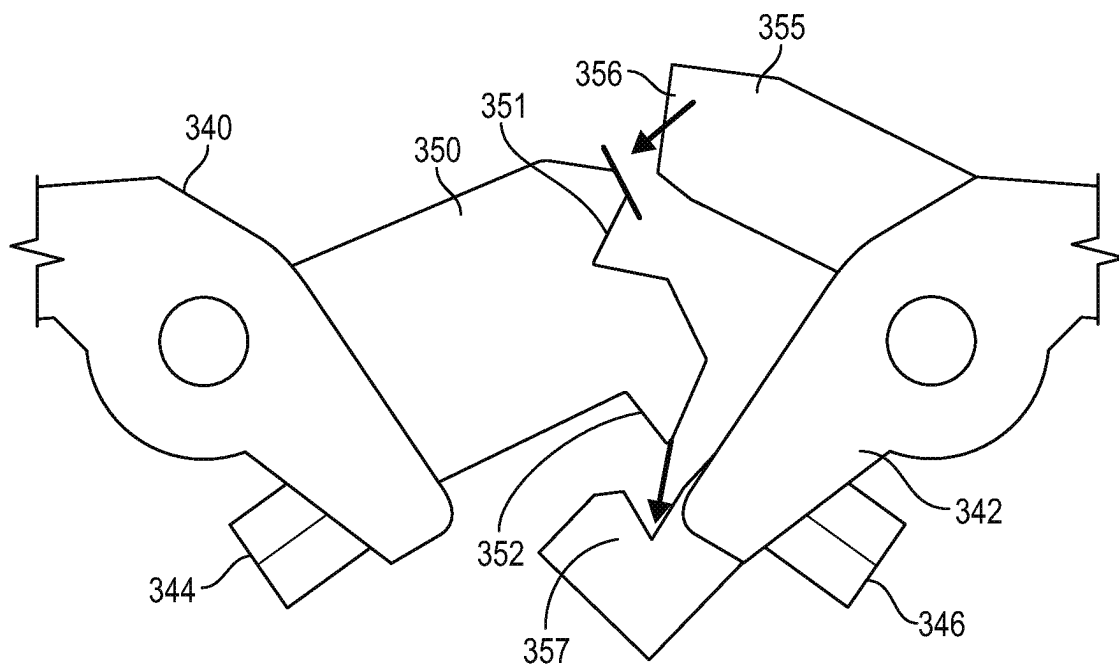


FIG. 22

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LOCKING MECHANISM USED WITH A FLAT WEBBING

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation-in-part application which claims priority to U.S. Non-Provisional application Ser. No. 17/685,152 filed on Mar. 2, 2022, which claims priority to U.S. Provisional Patent Application No. 63/155,739 filed on Mar. 3, 2021, which are incorporated by reference in their entirety.

FIELD OF THE DISCLOSURE

The present invention is directed to locking devices, and more particularly to a locking device that can be integrated onto a flat webbing.

BACKGROUND

Dog collars of various types exist. Some collars have a loop that passes over the dog's head and then is tightened. These collars generally take the form of "choker" collars that have no mechanism to stop the collar from tightening or loosening and are held on the dog's neck by keeping tension on the collar. Choker collars can restrict the breathing of a dog wearing the collar with enough tension. Further, if no tension is kept on the choker collar the collar may become loose and move out of position (e.g., over the head of the dog or tangled in the legs of the dog) or even fall off the dog.

Some collars exist that slip over the head of the dog and include mechanisms to stop the collar from becoming too tight. Others include mechanisms that prevent the collar from becoming too loose. However, all of these attempts have been inconvenient for a variety of reasons. One example of these inconvenient solutions is a collar with a strap that includes holes through which a locking mechanism may pass through to keep the collar from tightening or loosening. This solution is inconvenient because a person putting on the collar must place the locking mechanism in a hole while the collar is around the dog's neck, which may be difficult when a dog is excited to go outside or otherwise is not being compliant, and which is far less convenient than a choker collar which simply slips over the dog's head and is ready to use.

Another example of an inconvenient solution is a collar with a disc or similar device on the strap to prevent the collar from becoming too tight. Although many of these collars can be easily slipped over the head of the dog, the disc does not prevent the collar from becoming too loose or falling off the dog. Also, it can be difficult or impossible to adjust the collar size on the dog because the disc must be fixed to a location on the strap to prevent the collar from tightening.

SUMMARY

One or more embodiments are provided below for a locking device that may be used with a flat webbing material, such as may be used for a leash. The locking device may be integrated onto a flat webbing material to form a slip-on leash for a dog or another animal. In other words, the locking device integrated onto a webbing material may form a leash and collar all in one and that with the locking device the collar can be set to a certain width and then locked to prevent overtightening or loosening. The locking device may also be included on a dog collar that can be slipped around a dog's

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neck or another animal's neck. The locking device may also be on any other object that can benefit from such a locking device. The locking device may include a main body that is divided into a first section and a second section. One or more arms are connected within the second section. The locking device may also include a locking piece and a hook. The locking piece is configured to fit within the first section of the main body and includes a tab that projects outward from the locking piece which engages with an overhanging portion on one of the arms of the one or more arms to securely lock the locking piece within the main body. The hook is configured to fit within the second section with the one or more arms holding the hook in place within the second section of the main body. The one or more arms may be moved by pressing on to release the locking piece, the hook, or both.

The disclosed device is unique and may provide significant advantages when compared with other devices and solutions known in the arts because it provides a locking device that may be used with a slip-on dog leash or dog collar which prevents the device from tightening or loosening as the leash or collar is being pulled on. The disclosed device also provides a convenient method for putting or removing a slip-on dog leash or dog collar. Additionally, the locking device, when included on a slip-on dog leash or dog collar, allows for convenient adjustments in size of a collar portion of a strap of the device. Such a device may also be integrated onto a flat webbing material that can be used to wrap around an object, in the same manner as discussed above, to provide a possible tying means.

Other advantageous features as well as other aspects and advantages of the invention will be apparent from the following description and the appended claims.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments of the present disclosure are described in detail below with reference to the following drawings. These and other features, aspects, and advantages of the present disclosure will become better understood with regard to the following description, appended claims, and accompanying drawings. The drawings described herein are for illustrative purposes only of selected embodiments and not all possible implementations and are not intended to limit the scope of the present disclosure.

FIG. 1 shows an example perspective view of a locking device integrated onto a leash according to one or more non-limiting embodiments.

FIG. 2 shows a close-up view of collar section of the leash with the locking device according to one or more non-limiting embodiments.

FIG. 3A shows an exploded view of the locking device according to one or more non-limiting embodiments.

FIG. 3B shows a close-up view of the locking device in an engaged position according to one or more non-limiting embodiments.

FIG. 4 shows an example top view of a main body of the locking device according to one or more non-limiting embodiments.

FIG. 5A shows an example of a lock piece of the locking device according to one or more non-limiting embodiments.

FIG. 5B shows the lock piece and the leash before the lock piece is fixed to the leash according to one or more non-limiting embodiments.

FIG. 5C shows the lock piece fixed onto the leash and further shows a projecting element on the lock piece according to one or more non-limiting embodiments.

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FIG. 6A shows a partial cross-sectional view of the locking device in FIG. 3B taken along line B-B illustrating a hook engaging with a first and second arm of the locking device.

FIG. 6B shows the locking device from FIG. 6A illustrating the hook engaged with the first and second arms of the locking device.

FIG. 6C shows the locking device from FIG. 6B illustrating the hook disengaged from the first and second arms of the locking device.

FIG. 7A shows a partial cross-sectional view of the locking device in FIG. 3B taken along line A-A illustrating the lock piece engaging with the first arm.

FIG. 7B shows the locking device from FIG. 7A illustrating the lock piece disengaged from the first arm.

FIG. 8 is an example top view of the locking device according to one or more non-limiting embodiments.

FIG. 9A shows an example view of the collar portion of the leash with the lock piece disengaged from the main body of the locking device.

FIG. 9B shows the collar portion of the leash integrated with the locking device placed around a dog's neck with the lock piece disengaged from the main body.

FIG. 9C shows the locking device on the leash in a locked position.

FIG. 9D shows the hook released from the main body to quickly remove the collar portion of the leash from the dog's neck.

FIG. 10A shows an example view of an alternative embodiment of a locking device.

FIG. 10B shows the alternative embodiment of the locking device integrated with a leash and with a hook and a lock piece engaged in a main body of the locking device.

FIG. 11 is an exploded view of the alternative embodiment of the locking device.

FIG. 12 shows a top view of a main body of the alternative embodiment of the locking device.

FIG. 13 shows an internal view of the alternate embodiment of the locking device illustrating the hook engaged with the first and second arms.

FIG. 14 shows an internal view of the alternate embodiment of the locking device illustrating the lock piece engaged with the first and second arms.

FIG. 15 shows a perspective view of yet another alternative embodiment of a locking device.

FIG. 16 shown an exploded view of the alternative embodiment of the locking device from FIG. 15.

FIG. 17 shows a top perspective view of a main body of the alternative embodiment of the locking device of FIG. 15.

FIG. 18 shows a top view of the main body and a first arm and a second arm of the locking device of FIG. 15.

FIG. 19 shows a front perspective of the a lock piece of the locking device of FIG. 15.

FIG. 20 is a cross-sectional view of the locking device taking along line A-A from FIG. 15.

FIG. 21 is a cross-sectional view of the locking device taking along line B-B from FIG. 15.

FIG. 22 illustrates the first arm and the second arm with their respective projections.

DETAILED DESCRIPTION

In the Summary above, in this Detailed Description, the claims below, and in the accompanying drawings, reference is made to particular features of the invention. It is to be understood that the disclosure of the invention in this specification includes all possible combinations of such

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particular features. For example, where a particular feature is disclosed in the context of a particular aspect or embodiment of the invention, or a particular claim, that feature can also be used—to the extent possible—in combination with and/or in the context of other particular aspects and embodiments of the invention, and in the invention generally.

The term “comprises” and grammatical equivalents thereof are used herein to mean that other components, ingredients, steps, etc. are optionally present. For example, an article “comprising” (or “which comprises”) components A, B, and C can consist of (i.e., contain only) components A, B, and C, or can contain not only components A, B, and C but also contain one or more other components.

Where reference is made herein to a method comprising two or more defined steps, the defined steps can be carried out in any order or simultaneously (except where the context excludes that possibility), and the method can include one or more other steps which are carried out before any of the defined steps, between two of the defined steps, or after all the defined steps (except where the context excludes that possibility).

The term “at least” followed by a number is used herein to denote the start of a range including that number (which may be a range having an upper limit or no upper limit, depending on the variable being defined). For example, “at least 1” means 1 or more than 1. The term “at most” followed by a number is used herein to denote the end of a range, including that number (which may be a range having 1 or 0 as its lower limit, or a range having no lower limit, depending upon the variable being defined). For example, “at most 4” means 4 or less than 4, and “at most 40%” means 40% or less than 40%. When, in this specification, a range is given as “(a first number) to (a second number)” or “(a first number)-(a second number),” this means a range whose limits include both numbers. For example, “25 to 100” means a range whose lower limit is 25 and upper limit is 100 and includes both 25 and 100. Certain terminology and derivations thereof may be used in the following description for convenience in reference only and will not be limiting. For example, words such as “upward,” “downward,” “left,” and “right” would refer to directions in the drawings to which reference is made unless otherwise stated. Similarly, words such as “inward” and “outward” would refer to directions toward and away from, respectively, the geometric center of a device or area and designated parts thereof. References in the singular tense include the plural, and vice versa, unless otherwise noted.

The term “coupled to” as used herein may mean a direct or indirect connection via one or more components.

Referring now to the drawings and the following written description of the present invention, it will be readily understood by those persons skilled in the art that the present invention is susceptible to broad utility and application. Many embodiments and adaptations of the present invention other than those herein described, as well as many variations, modifications, and equivalent arrangements will be apparent from or reasonably suggested by the present invention and the detailed description thereof, without departing from the substance or scope of the present invention. This disclosure is only illustrative and exemplary of the present invention and is made merely for purposes of providing a full and enabling disclosure of the invention.

FIG. 1 shows a perspective view of an example of a non-limiting embodiment of a locking device 100 integrated onto a leash 400, which is a type of flat webbing, and that is adapted to be slipped onto an animal's neck, such as a dog. The locking device 100 can be integrated onto any flat

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webbing material and can be used together with the locking device 100 to wrap around an object. For the remainder of this disclosure, the leash 400 will be used as an example of a webbing material that is integrated with the locking device 100. Also, for the remainder of this disclosure, the leash 400 and locking device 100 will be used with a dog, as an example. The locking device 100 is integrated onto a leash 400 to form a collar section 410 and a strap section 420. The collar section 410 of the leash 400 forms a loop that is placed around the dog's neck. The collar section 410 further includes an adjustment piece 412, such as and not limited to a buckle or a slide glider, that allows the collar section 410 to widen or shorten. The strap section 420 of the leash may include a handle 422. The leash 400 may be similar to other leashes that are known and available in the arts or will be in the future. The leash 400 integrated with the locking device 100 can come in many sizes and lengths to accommodate small, medium, and large sized dogs. The leash 400, for example, may have a length of about 5-8 feet. In some example embodiments, the leash 400 may be parts of a single continuous webbing strap, cord, chain, rope, or other similar material. In other embodiments, portions of the leash 400 may be made of different materials.

The handle 422 and the collar portion 410 are formed on opposite ends of the leash 400 with a length of the strap 420 separating the two. The leash 400 may have a first end 401 and a second end 402. The collar portion 410 is formed near the first end 401. The handle 422 is formed near the second end 402, which may be formed by doubling over the strap 400 and attaching (by sewing, glue, rivet, etc.) the second end 402 of the leash 400 to another portion of the strap 422 to form a loop sized for a human hand. The adjustment piece 412 is attached to the leash 400 within the collar section 410. As mentioned above, the adjustment piece 412 allows the collar section 410 to widen or shorten the loop to fit around a dog's neck. It is to be understood that this is an example of how the leash 400 may be formed, however, there are variations to other forms and shapes that the leash 400 may take.

Refer to FIGS. 2, 3A, and 3B regarding the locking device 100 integrated on to the leash 400. FIG. 2 is close-up view of the collar section 410 of the leash 400 with the locking device in an unengaged position. FIG. 3A is an exploded view of the locking device 100. FIG. 3B is a close-up view of the locking device 100 in a fully engaged position. As shown in FIG. 2, the locking device 100 is integrated onto the leash 400 and the placement of the different elements of the locking device 100 assist in forming the collar section 410 of the leash 400. The locking device 100 comprises of a main body 110, a hook 120, a lock piece 130, and one or more arms 140, 142. The hook 120 is connected to the first end 401 of the leash 400. The lock piece 130 is fixedly connected to the leash 400 and is positioned between the adjustment piece 412 and the main body 110 which is positioned on the leash 400 between the lock piece 130 and the handle 422. The adjustment piece 412 of the leash is between the hook 120 and the lock piece 130. In other words, when looking at the leash 400 from the first end 401 to the second end 402, the hook 120 is connected to the first end 401, next is the adjustment piece 412 on the leash 400, next is the locking piece 130 which is fixed onto the leash, then the main body 110 of the locking device 100, then the strap section 420, and lastly the handle 422 configured from looping the second end 402.

FIG. 2 illustrates that the lock piece 130 is fixedly connected to the leash 400, and the leash 400 may then pass through a slit 112a (shown in FIG. 4) in the main body 110

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on a side that moves toward the second end 402 which has the handle 422. From the lock piece 130, on a side moving toward the first end 401, the leash 400 may pass through the adjustment piece 412 (with the fixed lock piece 130 positioned between the main body 110 and the adjustment piece 412). From the adjustment piece 412, the leash 400 may then pass through an opening 122 in the hook 120 of the locking device 100 and loop back to the adjustment piece 412. Because a section of the collar portion 410 between the hook 120 and the adjustment piece 412 is a looped section 414, the adjustment piece 412 can be used to adjust the length of the collar portion 410, which will be discussed below.

The strap portion 420 of the leash 400 may slide through the slit 112a (See, FIG. 4) in the main body 110 of the locking device 100, except when the lock piece 130 is engaged with the main body 110, which will be discussed in more detail below. In other words, the main body 110 is not fixed to the leash 400 such that the main body 110 may slide along the strap portion 420 between the handle and the lock piece 130. When the lock piece 130 is engaged with the main body 110, the main body 110 is prevented from sliding on the leash 400. When the lock piece 130 is not engaged with the main body 110, the collar portion 410 may be widened allowing for the collar portion 410 to be placed over the head of a dog. Then, the lock piece 130 may be slid toward the main body 110 by pulling on the strap section 420 to engage the lock piece 130 with the main body 110 to lock and prevent further movement of the main body 110 on the leash 400 and lock a size of the collar portion 410. The one or more arms 140, 146 are connected to the main body 110 and may be used to engage and disengage the lock piece 130 from the main body 110. Additionally, the one or more arms 140, 142 may also be used to engage and disengage the hook 120 from the main body 110. The hook 120 when engaged is attached to the main body 110, and when disengaged can be removed from the main body 110, allowing the collar portion 410 of the leash 400 to fully open and undo the loop of the collar portion 410. Removing the hook 120 from the main body 110 allows the collar portion 410 to come off the dog's neck quickly.

The size of the loop of the collar portion 410 may be adjusted using the adjustment piece 412. The looped section 414, and a straight section 416 (a section between the lock piece 130 and the adjustment piece 412) together determine the length of the loop of the collar portion 410 (when the lock piece 130 is engaged with the main body 110). By moving the adjustment mechanism 412, the looped section 414 can be lengthened or shortened, and this changes the size of the loop of the collar portion 410. The size of the loop of the collar portion may be increased (and conversely the length of the looped section 414 can be shortened) when the adjustment mechanism 412 is slid away from the lock piece 130 and toward the hook 120 of the locking device 100. The size of the loop of the collar portion 410 may be decreased (and conversely the length of the looped section 414 may be increased) when the adjustment piece 412 is moved toward the lock piece 130 and away from the hook 120 of the locking device 100.

Referring specifically to the locking device 100, FIGS. 3A and 3B illustrate the elements that comprise the locking device 100 and how the key components are connected to each other to provide a useful functionality. Also referring to FIG. 4, a top view of a non-limiting example of the main body 110 is shown. The main body 110 provides a structural support for the remainder of the elements comprising the locking device 100. The main body 110 is configured with four sides connected to a closed bottom end and having an

open top end. The terms “bottom” and “top” are strictly being used for reference purposes. The main body 110 is further configured with a first section 112 and a second section 114 separated by a center wall 116. The main body may be fashioned from a durable material including and not limited to plastic, metal, a composite, or other durable materials suited for withstanding the forces exerted by a dog pulling on a leash attached to the locking device 100.

The first section 112 and the second section 114 are configured to accept the lock piece 130 and the hook 120, respectively. The first section 112 has the slit 112a configured on the bottom end of the locking device 100. As mentioned above, the lock piece 130 is fixedly connected to the leash 400 and the slit 112a is adapted to allow the leash 400 to pass through when the locking device is attached to the leash 400. In operation, the leash 400 can slide back and forth through the slit 112a as the main body 110 is not fixed to the leash 400. In other words, the main body 110 is connected to the leash 400 by the leash 400 being inserted through the slit 112a. The leash 400 will slide through back and forth till the lock piece 130 slides into the first opening 112 of the main body 110 and is locked into place. The slit 112a has a width that allows the leash 400 to slide back and forth but not wide enough for the lock piece 130 to go through. In other words, the lock piece 130 abuts against the bottom end of the main body 110 which includes the slit 112a and allows the lock piece 130 to nest in first section 112 of the main body 110. The nesting of the lock piece 130 in the first section 112 of the main body 110 keeps the collar portion from tightening any further, so the width of the collar portion is set when the lock piece 130 is nesting in the main body 110.

The second section 114 is configured to include the one or more arms 140, 142. The one or more arms 140, 142 are attached to the main body 110 within the second section 114. As clearly seen in FIG. 3A, the second section 114 is configured with a depression 115 on the side defining part of the second section and which is parallel to the center wall 116. The depression 115 is included to allow the hook 120 to be inserted into the second section 114 and sit flush against the top side and further prevent any side-to-side movement of the hook 120, as shown in FIG. 3B. The second section further includes openings 117 that traverse the side of the main body 110 which defines part of the second section 114 and includes openings 117 in the center wall 116. The openings 117 on the side of the main body 110 and the center wall 116 are aligned together on the same plane so that the one or more arms 140, 142 may be to be connected to the main body 110 within the second section 114, which will be discussed below.

The first section 112 and the second section 114 of the main body 110 may be rectangular in shape. A rectangular shape is well suited for the leash 400 integrated with the lock device 130 that includes a flat shape for the leash 400. The first section 112 and second section 114 may have other shapes if the leash that passes through the first section 112 is a rope, chain, cable, etc. It is to be understood that the first section 112 and the second section 114 may have different shapes wherein the second section 114 may have a rectangular shape and the first section may have an alternate shape designed to accommodate the shape of the leash 400.

The lock piece 130 is depicted in FIGS. 3A-B and 5A-C. In FIG. 3A, the lock device 130 is shown to be integrated onto the leash 400 which is inserted into the slit 112a in the first section 112 of the main body 110. In FIG. 3B, the lock device is inserted into the first section and engaged with the one or more arms 140, 146, which will be discussed below.

In FIGS. 5A-5C, the lock device 130 is shown to be comprised of two pieces, a first lock piece 130a and a second lock piece 130b which are fixed onto the leash 400 with the leash 400 between the first and second lock pieces 130a, 130b. As best seen in FIG. 5A, an inside surface of the first and second lock pieces 130a, 130b are shown. FIG. 5B shows that the inside surfaces of the first and second lock pieces 130a, 130b, face each other and attach to the leash 400. In some embodiments, the inside surfaces of the first and second lock pieces 130a, 130b are configured with a plurality of spikes 132, 134, respectively. One example of the spikes 132, 134, may be an arrowhead type appearance with a pointed tip. The spikes are useful to pierce into the leash 400 to partially secure the lock piece 130 to the leash 400. It is to be understood that the first and second lock pieces 130a, 130b, may not include the spikes 132, 134.

The first and second lock pieces 130a, 130b are attached to the leash 400 by using a securing element. Referring to FIGS. 5A-C, in one or more embodiments, the first and second lock pieces 130a, 130b, may have a plurality of holes 136 that traverse through each lock piece 130. The holes 136 on the first lock piece 130a and the second lock piece 130b align with each other when each lock piece 130 is placed onto the leash 400. The plurality of holes 136 allow the first and second lock pieces 130a and 130b to be fixed to the leash 400 and to each other. The plurality of holes 136 are particularly useful for using a fastener 137 to be placed through to attach the lock pieces 130 together to the leash 400. An example of a fastener 137 may include rivets. It is also to be understood that the first and second pieces 130a, 130b, may be fixed to each other and the leash 400 by using other securing elements such as and not limited to thread, glue, staples, or another securing element.

FIG. 5C shows an example side view of the lock device 130 on the leash 400. In this Figure, only the first lock piece 130a of the strap 100 is shown to allow for greater detail to be shown on the described portion. The first lock piece 130a has a projecting element 138 on an outward facing surface (surface opposite the side fixed to the leash 400) which is shaped to engage with the one or more arms 140, 142 to lock the locking device 130 into the main body 110. The first lock piece 130a with the projecting element 138 when slid into the first section 112 of the main body 110 enters facing the center wall 116. The center wall 116 has a groove 118 that is open from the top end of the main body 110 to the bottom end of the main body 110. The projecting element 138, as the name suggests, projects outward from the first lock piece 130a, and as the lock device slides into the first element 112, the projecting element is guided into the groove 118 of the center wall 116 (See, FIG. 7A).

As discussed earlier, the one or more arms 140, 142, are connected to the main body 110 within the second section 114. The one or more arms 140, 142, may be configured to cause the lock piece 130 to engage and disengage within the main body 110 of the locking device 100. Additionally, the one or more arms 140, 142, may also be configured to cause the hook 120 to engage and disengage within the main body 110. As seen in the Figures, the locking device 100 has two arms, a first arm 140 and a second arm 142. Each of the first and second arms 140, 142, are attached to the main body 110 within the second section 114 by a pin 145 which is inserted into the openings 117 in the second section and the openings 141, 143, on the first and second arms 140, 142, respectively. The pin 145 fixes each of the first and second arms 140, 142 to the first section 114 of the main body. A portion of the first and second arms 140, 142 extends outside of the main body 110, which allows a surface for a user to press on to move

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the first and second arms **140**, **142** at the pin **145** which functions as an axel. Stated another way, the first and second arms **140**, **142** rotate at the pin **145** within the second section **114** of the main body **110**.

The first and second arms **140**, **142** extend toward the bottom end of the main body **110** and a tip of each arm **140**, **142** at the bottom end are relatively close to each other or almost touching. Referring to FIG. 7A the first and second arms **140**, **142**, may each include a spring engaging portion **140a**, **142a**, respectively. Specifically, the spring engaging portions **140a**, **142a** are relatively configured near the tip of each arm **140**, **142**, and facing toward the side of the main body **110**. Each of the spring engaging portions **140a**, **142a**, are attached to a spring device **144**, **146**, respectively, which push against an inside wall of the second section **114** in the main body **110**. The spring engaging portions **140a**, **142a** attached to their respective spring device **144**, **146**, provide a restoring force on the first and second arms **140**, **142**, respectively. A user applies pressure (by pressing) on the first and second arms **140**, **142**, rotating the first and second arms at the pin **145**. The rotation with the applied pressure moves the first and second **140**, **142**, upward at the outside portion and the tips away from a center of the bottom end of the second section **114**. Releasing the pressure off the first and second arms **140**, **142**, returns the first and second arms **140**, **142**, back to their resting position which is with the tips near the center of the bottom end of the second section **114** of the main body **110**. The spring devices **144**, **146** may be steel springs or other similar devices that provide a restoring force. The first and second arms **140**, **142** may be fashioned from the same material as the main body **110** including and not limited to metal, plastic, or other durable materials.

FIGS. 6A to 6C is a cross-sectional view taken along line B-B in FIG. 3B to depict the hook **120** engaging and disengaging with the first and second arms **140**, **142**. Disengaging the hook **120** detaches the hook from the main body **110** and opens up the collar portion **410**. In other words, the detaching the hook **120** from the main body undoes the collar portion **410**. The hook **120** has a broad tip **124** which enters the second section **114** and engages with the first and second arms **140**, **142**. FIGS. 6A and 6B show that as the hook **120** is inserted into the second section **114**, the broad tip **124** pushes against the first and second arms **140**, **142**, causing them to move backwards to make space for the broad tip **124** on the hook **120** to move past the first and second arms **140**, **142** to fit at the bottom end of the second section **114**. Once the broad tip **124** on the hook **120** moves past the first and second arms **140**, **142**, the spring devices **144**, **146** move the first and second arms **140** back to their original position engaging them over the broad tip **124** and preventing the hook **120** from moving. Thus, the first and second arms **140**, **142** lock the hook **120** into the second section **114**. FIG. 6C shows that the hook **120** is disengaged and unlocked when the first and second arms **140**, **142** are moved at the pins **145** causing the first and second arms **140**, **142** to move away from the broad tip **124** freeing the hook **120** to slide out of the second section **114**.

FIGS. 3A and 8 also illustrate that the first arm **140** has a wide tip **148** which extends toward the center wall **116**, and more particularly toward the groove **118** in the center wall **116**. The wide tip **148** locks the lock piece **130** in the first section **112** of the main body **110**. Specifically, the wide tip **148** of the first arm **140** engages with the projecting element **138** on the lock piece **130** to hold the lock piece **130** within the first section **112** of the main body **110**. FIGS. 7A and 7B show example cross-sectional views taken from FIG. 3B along line A-A. FIG. 7A and FIG. 8 show the first and

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second arms **140**, **146** connected within the second section **114** with a view of the lock device **130** within the first section (not shown in this figure). The projecting element **138** is shown in the groove **118** of the center wall **116**. The first arm **140** is shown to be engaged with the lock piece **130** by preventing the movement of the lock piece out of the first section of the main body **110**. Specifically, the wide tip **148** is above the projecting element **138** in the groove of the main body **110** such that the lock piece **130** is prevented from moving upward by the projecting element **138** prevented from moving upward in the groove **118** by the wide tip **148**. FIG. 7B shows the first arm **140** disengaged from the lock piece **130**. Specifically, the first arm **140** may be moved at the pin **145** causing the wide tip **148** to be moved away from a position above the projecting element **138** and thus allowing the lock device **130** to move upward with the projecting element **138** moving upward through the groove **118** unhindered.

FIGS. 9A to 9D depict an embodiment of the locking device **100** integrated onto a leash **400** being used with a dog **500**. In FIG. 9A, a user would ensure that the lock piece is disengaged and not inserted into the main body **110** of the locking device. The user may widen the collar portion **410** of the leash **400** to be able to slip it over the dog's head. FIG. 9B shows that the collar portion **410** has been slipped over the dog's head and is loosely placed around the dog's neck. In FIG. 9C, the strap section **420** of the leash **410** may be pulled on to slide the leash through the main body **110** moving the lock piece **130** into the main body **110** to engage with the first arm locking the lock piece into place. Once locked, the width of the collar portion **410** is set and as the leash is pulled on while around the dog's neck, the collar portion will not tighten or loosen. FIG. 9D shows where the hook **120** can detach from the main body **110** for a quick release of the locking device from the dog without needing to slip the collar portion **410** over the dog's head. This may be useful for speedily releasing the dog from the leash **400**.

The locking device **100** integrated onto a leash **400** has several advantages over existing dog collars. The locking device **100** may be slipped over the head of the dog and secured by sliding the main body **110** along the leash until the lock piece **130** fixed onto the leash **400** enters the main body **110**. Thus, the locking device **100** integrated onto a leash may be more easily and more conveniently placed over on a dog. The lock piece **130** engages with the first arm to prevent the collar portion **410** placed around the dog's neck from loosening by preventing the lock piece from sliding back out of the open top end. Additionally, the lock piece **130** is able to nest within the main body by pushing against the closed bottom end and thus the collar portion is prevented from tightening a further on the dog's neck. The size of the collar portion **410** can be adjusted using the adjustment mechanism **412** so the leash **400** integrated with the locking device **100** can be adjusted for dogs of different sizes. The quick release mechanism of the hook **120** allows the leash **400** integrated with the locking device **100** to be quickly and easily released from around the dog's neck.

FIGS. 10A to 14 depict an alternative embodiment of a locking device **200** that may be integrated onto a leash **400** that can be secured around the neck of a dog similar to the disclosure above in reference to the first embodiment of the locking device **100** of FIG. 1. FIG. 10A illustrates the locking device **200** only in order to provide a clear view of all the elements that comprise the locking device **200**. FIG. 10B illustrates the locking device **200** integrated with leash **400** to show that the like elements of the locking device **200** are integrated onto the leash **400** as discussed above in

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reference to the locking device 100 integrated onto the leash 400. FIG. 11 illustrates an exploded view of the locking device 200 only to provide a clearer view of the elements comprising the locking device 200.

The locking device 200 comprises of similar elements to the locking device 100 including a main body 210, a hook 220, a lock piece 230, and first and second arms 240, 242. All elements are the same as discussed in the above embodiment of the locking device 100. There are a few differences which will be discussed herein. As best seen in FIGS. 11 and 12, the main body 210 has a center wall 216 with two parallel grooves 218 separated by a center post 216a (whereas the locking device 100 discussed above has one groove 118). The first and second arms 240, 242 both engage with the lock piece 230, and both the first and second arms 240, 242 are configured with a wide tip 248, 249, respectively as seen in FIGS. 11 and 14. Thus, as seen in FIG. 14, the lock piece 230 is configured with two projecting elements 238 wherein each of the projecting elements is guided into one groove so that each of the two grooves has one of the projecting elements 238 side by side as the lock piece is inserted into the first section. The two projecting elements 238 engage with the wide tips 248, 249, on the first and second arms 240, 242, respectively. The first and second arms 240, 242, also engage with the hook 220. The process to engage and disengage the hook 220 is the same as discussed above for the locking device 100. However, both the first and second arms 240, 242, need to be pressed on to move the arms 240, 242 away from the projecting elements 238 on the lock piece to allow the lock to be moved out of a first section 212 of the main body 210. The first and second arms 240, 242 need only be pressed halfway to move them away from the projecting elements 238 to disengage and move the lock piece 230. Fully depressing the first and second arms 240, 242, moves them away from a broad tip 224 of the hook 220 and thus allowing both the hook 220 and the lock piece 230 to disengage and move out of the main body 210.

FIGS. 15 to 22 depict yet another alternative embodiment of a locking device 300 that may be integrated onto a leash 400, as shown in the previous two embodiments. This non-limiting embodiment of the locking device 300 can be used with the leash 400, similar to the disclosure above in reference to the first two embodiments of the locking device 100, 200 of FIG. 1 and FIG. 10, respectively. FIG. 15 is a side perspective view of the locking device 300 to illustrate how elements comprising the locking device 300 are put together. The locking device 300 is not shown integrated to a leash 400, however, the locking device 300 would be integrated with a leash similarly to how the locking devices 100, 200 are integrated with their respective leash 400. The integration of the leash 400 is discussed above in reference to the locking devices 100, 200. FIG. 16 illustrates an exploded view of the locking device 300 to provide a clearer view of the elements comprising the locking device 300.

The locking device 300 comprises of similar elements to the locking device 200 including a main body 310, a hook 320, a lock piece 330, and first and second arms 340, 342. All elements are the same as discussed in the above embodiment of the locking devices 100, 200 discussed in FIGS. 1 to 14. Like numbers denote like elements which include the same feature and functionality. There are a few differences which will be discussed herein. The main body 310 has a center wall 316 with a central groove 318, which includes a knob 316a protruding from a relative center of the central groove (whereas the locking device 200 has the central groove 218 with a post 216a extending from the center of the

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central groove 216; and whereas the locking device 100 discussed above has one groove 118).

As seen in FIG. 19, the lock piece 330 comprises of two pieces, a first lock piece 330a and a second lock piece 330b, wherein the first lock piece includes a projecting element 338 with a channel 338a on a bottom side of the projecting element. The projecting element 338 slides into the central groove 318 of the main body 310 and the channel 338a is guided onto the knob 316a which ensures that the lock piece 330 is positioned properly into a first section 312 of the main body 330.

The first and second arms 340, 342 are configured to engage with the lock piece 330 and the hook 320. The first and second arms 340, 342 are also configured such that both the first and second arms 340, 342 have reduced individual movement essentially keeping each from being pushed individually and only allowing the first and second arms 340, 342 to move together. As best seen in FIGS. 16 and 18, both the first and second arms 340, 342 are configured with a wide tip 348, 349, respectively, which are like the wide tips 248, 249 on the first and second arms 240, 242 of the locking device 200. . . . The first and second arms 340, 342 engage with the lock piece 330 by having the wide tips 348, 349 resting against a top of the projecting element 338. FIG. 21 is a cross sectional view of the locking device 300 taking along the line B-B to show the wide tips 348, 349 resting atop the projecting element 338.

The first and second arms 340, 342 also engage with the hook 320. The process to engage and disengage the hook 320 is the same as discussed above for the locking device 100. Once the broad tip 324 on the hook 320 moves past the first and second arms 340, 342 into the main body 310, spring devices 344, 346 on the first and second arms function similarly to the spring devices 144, 146 on the locking device 100. The spring devices 344, 346 move the first and second arms 340, 342 back to their original position engaging them over the broad tip 324 and preventing the hook 320 from moving. Similar to the second embodiment of the locking device 200, both the first and second arms 340, 342, need to be pressed on to move the arms 340, 342 away from the projecting element 338 on the first lock piece 330a to allow the lock piece 330 to be moved out of the first section 312 of the main body 310. The first and second arms 340, 342 need only be pressed halfway to move them away from the projecting element 338 to disengage and move the lock piece 330. Fully depressing the first and second arms 340, 342, moves them away from a broad tip 324 of the hook, allowing both the hook 320 and the lock piece 330 to disengage and move out of the main body 310. FIG. 20 illustrates a cross sectional view of the locking device taken along line A-A to illustrate the engagement of the first and second arms 340, 342 with the broad tip 324 of the hook 320.

However, as mentioned above, the first and second arms 340, 342 are configured to restrict individual movement and only allow them to move together. The first and second arms 340, 342 may allow a few degrees of individual movement such that the arms 340, 342 are pressed on simultaneously, they may move out of each other's way to fully disengage from the lock piece 330. As best seen in FIGS. 16, 18 and 22, the first arm 340 has a projection, referred to as a first arm projection 350, which extends relatively perpendicular from a top surface of the first arm and is substantially a length of the top surface. The first arm projection 350 also includes an inset portion 351, and a hook portion 352. The second arm 342 also includes a projection, a second arm projection 355, which extends perpendicularly from a top surface of the second arm 342, wherein the second arm

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projection 355 has an end 356 that is configured to catch into the inset portion 351 of the first arm projection 350. The second arm projection 355 also has an extension portion 357 at a bottom end of the second arm 342 which is configured to catch onto the hook portion 352 of the first arm projection 350. In this way, the first and second arm projections 350, 355 restrict the individual movement of the arms 340, 342 to only a few degrees, such as but not limited to, movement between 5 to 12 degrees.

Referring to FIG. 18, it can be seen that the first arm and second arm projections 350, 355 extend from a portion of the first arm 340 and a portion of the second arm 342 such that a remainder of the top surface of the first arm and a remainder of the top surface of the second arm do not have any projection and are smooth. When the first and second arms 340, 342 are placed within a second section 314 of the main body 310, the first and second arm projections 350, 355 abut against the center wall 316. Thus, these projections 350, 355 are not in the way of the hook piece 320 entering the second section 314 which would slide over the smooth portion of the first and second arms 340, 342 when inserted into the second section 314 of the main body 310. The hook piece 320 functions similar to the hook piece 120 described in the embodiment of the locking device 100. Thus, like the locking device 100 with the hook 120, the hook 320 of the current embodiment of the locking device 300 is inserted into the second section 314, a broad tip 324 pushes against the smooth portion of the first and second arms 340, 342, causing them to move backwards to make space for the broad tip 324 to move past the first and second arms 340, 342 to fit at the bottom end of the second section 314. However, the hook 320 has a hinge section 321, which is a section that rests in a depression 315 in the second section 314. The depression has a similar function as that described in the above embodiment of the locking device, which is included to guide the hook 320 to be inserted properly into the second section 314 and prevent side to side movement of the hook 320. The hinge section 321 reduces non-linear tension on the hook 320 and allows the hook 320 to move linearly as a connected leash is pulled on.

Accordingly, the present description provides for various embodiments for a locking device that can be integrated onto a flat webbing material that can be used to secure the flat webbing material around an object. An example use may include integrating the locking device onto a leash material that can be secured around the neck of a dog or other animal. Many uses and advantages are offered by the locking device as described above in one or more non-limiting embodiments in the present description.

The corresponding structures, materials, acts, and equivalents of any means or step plus function elements in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed. The description of the present invention has been presented for purposes of illustration and description but is not intended to be exhaustive or limited to the invention in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the invention.

The embodiments were chosen and described in order to best explain the principles of the invention and the practical application, and to enable others of ordinary skill in the art to understand the invention for various embodiments with various modifications as are suited to the particular use contemplated. The present invention, according to one or more embodiments described in the present description, may

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be practiced with modification and alteration within the spirit and scope of the appended claims. Thus, the description is to be regarded as illustrative instead of restrictive of the present invention.

What is claimed is:

1. A device comprising:

a main body having four sides, a center wall, a closed bottom, and an open top, wherein the center wall divides the main body into a first section and a second section;

a first arm and a second arm connected to the main body; a hook configured to slide into the main body and engage with the first arm and the second arm within the main body, wherein the hook has a hinged section;

a lock piece configured to slide into the main body to nest against the closed bottom and engage with the first arm and the second arm; and

wherein the hook and the lock piece are fixedly connected to a flat webbing material and the main body has a slit in the closed bottom of the first section, wherein the flat webbing material is inserted through the slit, so the main body is slidably connected to the flat webbing material, wherein the main body is prevented from sliding when the lock piece slides into the main body.

2. The device of claim 1, wherein the hook removably connects to the lock piece to form a collar portion and a strap with the flat webbing material, wherein an adjustment piece between the hook and the lock piece is configured to adjust a length of the collar portion.

3. The device of claim 1, wherein the hook has an opening configured to allow the flat webbing material to fit through and loop around the opening to connect the flat webbing material to the hook, and wherein the hinged section fits within a recessed portion of the main body.

4. The device of claim 1, wherein the lock piece has a first piece and a second piece which are fixedly connected to the flat webbing material, wherein the flat webbing material is positioned between and fixedly connected to the first piece and the second piece by one or more securing elements.

5. The device of claim 4, wherein a surface of the first piece and a surface of the second piece which are placed against the leash material are each configured with a plurality of spikes.

6. The device of claim 2, wherein the collar section is formed with the lock piece and the hook engaged with the first and second arms in the main body, and wherein the lock piece is nesting within the first section and the hook is inserted in the second section.

7. The device of claim 1, wherein the first arm and the second arm are connected to the main body by a pin which is placed within openings that traverse through the first and second arms, a side of the main body, and the center wall defining the second section, wherein the first and second arms are configured to move at the pin, and wherein the first and second arms each have a spring engaging portion and a connected spring for moving the first and second arms back to their original positions when they are moved at the pins.

8. The device of claim 1, wherein the hook is configured with a broad tip which engages with the first and second arms when the hook is inserted into the second section of the main body;

wherein the first arm and the second arm are engaged with the hook by resting above the broad tip to prevent the hook from sliding out of the second section; and

wherein the hook is disengaged when the first and second arms are moved away from the broad tip of the hook freeing the hook to slide out of the second section.

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9. The device of claim 6, wherein a lower end of the lock piece, facing toward a center wall, includes a projecting element having a channel, and

wherein the center wall has a central groove with a knob included on the closed bottom within the central groove, wherein the projecting element of the lock piece slides into the central groove with the knob accommodating the channel of the projecting element when the lock piece slides into the first section.

10. The device of claim 9, wherein the first arm and the second arm each have a wide tip wherein each of the wide tips extends toward the central groove in the center wall to engage with the projecting element, wherein the wide tips extending above the projecting element prevent the lock piece from moving out of the first section; and

wherein the lock piece is disengaged when the wide tips of the first arm and the second arm are moved away from the projecting element freeing the lock piece to slide out of the first section.

11. A device comprising:

a main body having a first section and a second section divided by a center wall, wherein a top end is open, and a bottom end is closed, and the first section has a slit in the bottom end;

a first arm and a second arm connected to the main body, wherein the first arm includes a first arm projection extending from a portion of a top surface of the first arm, and the second arm includes a second arm projection extending from a portion of a top surface of the second arm

a hook configured to slide into the main body to engage with the first arm and the second arm within the main body;

a lock piece configured to slide into the main body to nest against the closed bottom and engage with the first arm and the second arm; and

wherein the hook and the lock piece are fixedly connected to a flat webbing material, wherein the flat webbing material is inserted through the slit in the first section, so the main body is slidably connected to the flat webbing material, wherein the main body is prevented from sliding when the lock piece slides into the main body.

12. The device of claim 11, wherein:

the first arm projection extends from the portion of the top surface of the first arm such that the first arm projection abuts against the center wall and a remainder of the top surface of the first arm is smooth;

the second arm projection extends from the portion of the top surface of the second arm such that the second arm projection abuts against the center wall and a remainder of the top surface of the second arm is smooth; and

the first arm projection and the second arm projection engage with each other to restrict individual movement of the first arm and the second arm.

13. The device of claim 11, wherein the hook removably connects to the lock piece to form a collar portion and a strap with the flat webbing material, wherein an adjustment piece between the hook and the lock piece is configured to adjust a length of the collar portion.

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14. The device of claim 13, wherein the collar portion is formed with the lock piece and the hook engaged with the first and second arms in the main body, and wherein the lock piece is nesting within the first section and the hook is inserted in the second section.

15. The device of claim 11, wherein the hook has an opening configured to allow the flat webbing material to fit through and loop around the opening to connect the flat webbing material to the hook, and wherein the hook further includes a hinged section that fits within a recessed portion of the main body.

16. The device of claim 11, wherein the lock piece has a first piece and a second piece which are fixedly connected to the leash material, wherein the leash material is positioned between and fixedly connected to the first piece and the second piece by one or more securing elements, and

wherein a surface of the first piece and a surface of the second piece which are placed against the leash material are each configured with a plurality of spikes.

17. The device of claim 11, wherein the first arm and the second arm are connected to the main body by a pin which is placed within openings that traverse through the first and second arms, a side of the main body, and the center wall defining the second section, wherein the first and second arms are configured to move at the pin, and wherein the first and second arms each have a spring engaging portion and a connected spring for moving the first and second arms back to their original positions when they are moved at the pins.

18. The device of claim 12, wherein the hook is configured with a broad tip which slides over the flat section of the top surface of the first arm and the second arm when the hook is inserted into the second section of the main body;

wherein the first arm and the second arm are engaged with the hook by resting above the broad tip to prevent the hook from sliding out of the second section; and

wherein the hook is disengaged when the first arm and the second arm are moved away from the broad tip of the hook freeing the hook to slide out of the second section.

19. The device of claim 14, wherein a lower end of the lock piece, facing toward a center wall, includes a projecting element having a channel facing toward the closed bottom, and

wherein the center wall has a central groove with a knob included on the closed bottom within the central groove, wherein the projecting element of the lock piece slides into the central groove with the knob accommodating the channel of the projecting element when the lock piece slides into the first section.

20. The device of claim 19, wherein the first arm and the second arm each have a wide tip wherein each of the wide tips extends toward the central groove in the center wall to engage with the projecting element of the lock piece, wherein the wide tips extending above the projecting element prevent the lock piece from moving out of the first section; and

wherein the lock piece is disengaged when the wide tips of the first arm and the second arm are moved away from the projecting element freeing the lock piece to slide out of the first section.

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